

Physical Layer and Transmission Media

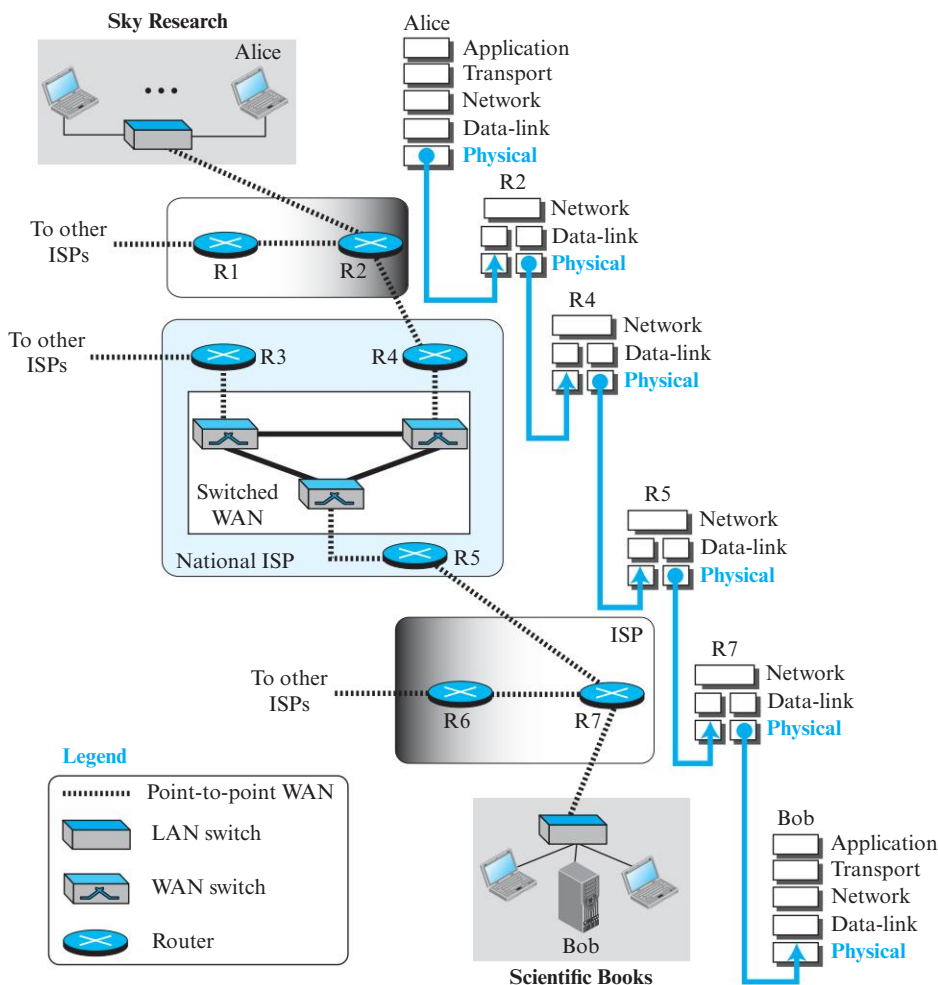
Our discussion of the TCP/IP protocol suite would be incomplete without the discussion of the physical layer and the transmission media that carry the signals created by the physical layer. However, this chapter can be partially or totally skipped if the reader has some knowledge about the physical layer. We should say that the role of the physical layer is to transfer the bits received from the data-link layer and convert them to electromagnetic signals for transmission. The first four sections of this chapter discuss this task. After the bits are converted to signals, the signals are delivered to the transmission media, which are the subject of our discussion in the last section. We have divided this chapter into five sections:

- ❑ In the first section, we first discuss the relationship between data and signals. We then show how data and signals can be both analog and digital. We then discuss analog signals and their characteristics and digital signals and their characteristics. We also discuss some issues related to the capacity and performance of physical channels.
- ❑ In the second section, we concentrate on digital transmission. We first show how to convert digital data to digital signals. We then show how to convert analog data to digital signals.
- ❑ In the third section, we concentrate on analog transmission. We first show how to convert digital data to analog signals. We then show how to convert analog data to analog signals.
- ❑ In the fourth section, we first show how multiplexing techniques can combine several low-bandwidth signals into a high-bandwidth signal, which allows us to use one transmission medium to carry several channels. We then show how spread spectrum techniques can safeguard a signal from intrusion.
- ❑ In the fifth section, we go below the physical layer and discuss the transmission media that are used in data communication. We introduce both guided (wires and cables) and unguided media (air).

7.1 DATA AND SIGNALS

At the physical layer, the communication is node-to-node, but the nodes exchange electromagnetic signals. Figure 7.1 uses the same scenario we showed in four earlier chapters, but the communication is now at the physical layer.

Figure 7.1 Communication at the physical layer



One of the major functions of the physical layer is to route bits between nodes. However, bits, as the representation of two possible values stored in the memory of a node (host, router, or switch), cannot be sent directly to the transmission medium (wire or air); the bits need to be changed to signals before transmission. So the main duty of the physical layer is to efficiently convert these bits into electromagnetic signals. We

first need to understand the nature of the data and then the types of signals to see how we can do this conversion efficiently.

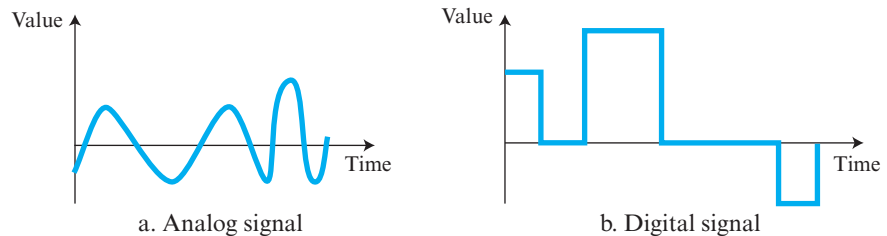
7.1.1 Analog and Digital

Data can be analog or digital. The term **analog data** refers to information that is continuous. Analog data, such as the sounds made by a human voice, take on continuous values. When someone speaks, an analog wave is created in the air. This can be captured by a microphone and converted to an analog signal or sampled and converted to a digital signal.

Digital data take on discrete values. For example, data are stored in computer memory in the form of 0s and 1s. They can be converted to a digital signal or modulated into an analog signal for transmission across a medium.

Like the data they represent, *signals* can be either analog or digital. An **analog signal** has infinitely many levels of intensity over a period of time. As the wave moves from value *A* to value *B*, it passes through and includes an infinite number of values along its path. A **digital signal**, on the other hand, can have only a limited number of defined values. Although each value can be any number, it is often as simple as 1 and 0. The simplest way to show signals is by plotting them on a pair of perpendicular axes. The vertical axis represents the value or strength of a signal. The horizontal axis represents time. Figure 7.2 illustrates an analog signal and a digital signal.

Figure 7.2 Comparison of analog and digital signals

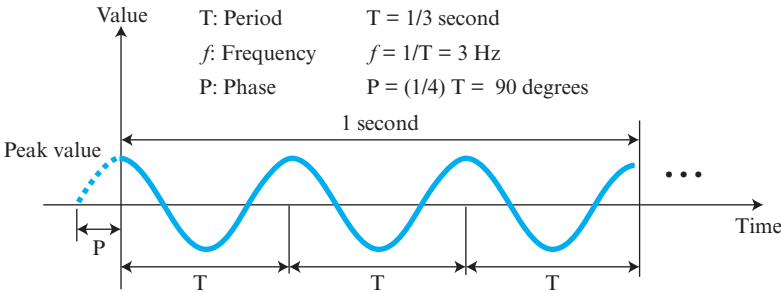


Analog Signals

Let us first concentrate on analog signals. An analog signal can take one of two forms: *periodic* or *nonperiodic (aperiodic)*. A **periodic analog signal** completes a pattern within a measurable time frame, called a **period**, and repeats that pattern over subsequent identical periods. The completion of one full pattern is called a *cycle*. A **nonperiodic analog signal** changes without exhibiting a pattern or cycle that repeats over time. In data communications, we commonly use periodic analog signals.

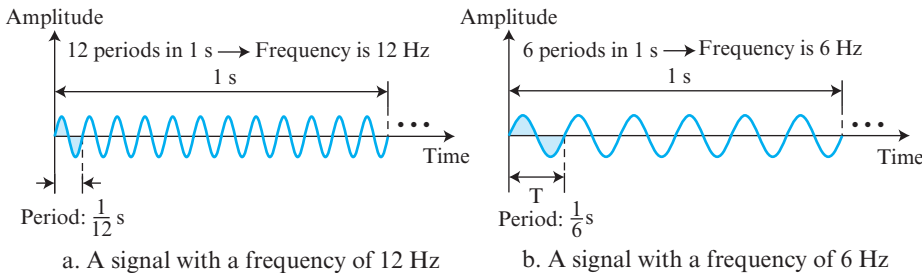
Periodic analog signals can be classified as simple or composite. A simple periodic analog signal, a **sine wave**, cannot be decomposed into simpler signals. A composite periodic analog signal is composed of multiple sine waves. The sine wave is the most fundamental form of a periodic analog signal. When we visualize it as a simple oscillating curve, its change over the course of a cycle is smooth and consistent, a continuous, rolling flow. Figure 7.3 shows a sine wave. Each cycle consists of a single arc above the time axis followed by a single arc below it.

Figure 7.3 A sine wave



A sine wave can be represented by three parameters: the *peak amplitude*, the *frequency*, and the *phase*. These three parameters fully describe a sine wave. The **peak amplitude** of a signal is the absolute value of its highest intensity, proportional to the energy it carries. For electric signals, peak amplitude is normally measured in volts. Figure 7.4 shows two signals and their peak amplitudes. *Period* refers to the amount of time, in seconds, a signal needs to complete 1 cycle. **Frequency**, measured in Hertz (Hz), refers to the number of periods in 1 second. Note that period and frequency are just one characteristic defined in two ways. Period and frequency are inverse of each other ($f = 1/T$).

Figure 7.4 Wavelength and period



The term **phase** describes the position of the waveform relative to time 0. If we think of the wave as something that can be shifted backward or forward along the time axis, phase describes the amount of that shift. It indicates the status of the first cycle. Phase is measured in degrees or radians (360° is 2π radian).

Wavelength is another characteristic of a signal traveling through a transmission medium. The wavelength is the distance a simple signal can travel in one period. Wavelength binds the period or the frequency of a simple sine wave to the propagation speed of the medium (see Figure 7.4).

While the frequency of a signal is independent of the medium, the wavelength depends on both the frequency and the medium. Wavelength is a property of any type of signal. In data communications, we often use wavelength to describe the transmission of light in an optical fiber.

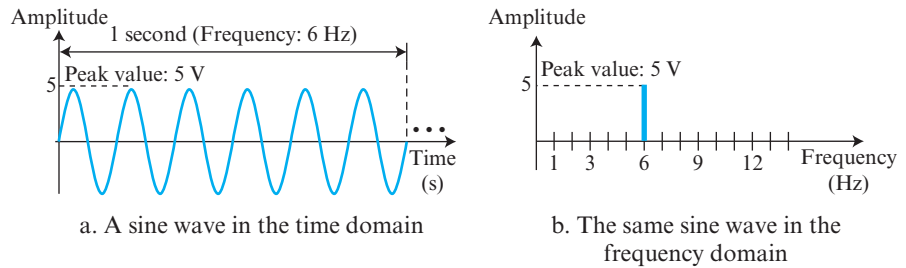
Wavelength can be calculated if one is given the propagation speed of the medium and the frequency of the signal. If we represent wavelength by λ , propagation speed by c (speed of light), and frequency by f , we get

$$\lambda = c / f = c \times T$$

Time and Frequency Domains

A sine wave is comprehensively defined by its amplitude, frequency, and phase. We have been showing a sine wave by using what is called a **time-domain plot**, which shows changes in signal amplitude with respect to time. To show the relationship between amplitude and frequency, we can use a **frequency-domain plot**. Figure 7.5 shows a signal in both the time and frequency domains.

Figure 7.5 The time-domain and frequency-domain plots of a sine wave

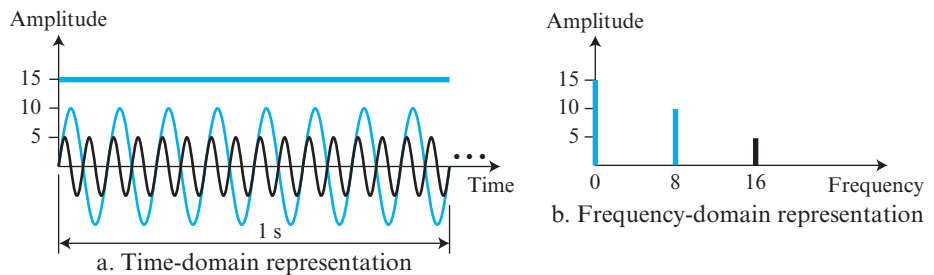


In the frequency domain, a sine wave is represented by one spike. The position of the spike shows the frequency; its height shows the peak amplitude.

Example 7.1

The frequency domain is more compact and useful when we are dealing with more than one sine wave. For example, Figure 7.6 shows three sine waves, each with different amplitude and frequency. All can be represented by three spikes in the frequency domain.

Figure 7.6 The time domain and frequency domain of three sine waves



Composite Signals

So far, we have focused on simple sine waves. Simple sine waves have many applications in daily life, such as sending energy from one place to another. However, if we had only one single sine wave to convey a conversation over the phone, it would make no sense and

carry no information. We would just hear a buzz; we need to send a composite signal to communicate data. A **composite signal** is made of many simple sine waves.

In the early 1900s, the French mathematician Jean-Baptiste Fourier showed that any composite signal is actually a combination of simple sine waves with different frequencies, amplitudes, and phases.

A composite signal can be periodic or nonperiodic. A periodic composite signal can be decomposed into a series of simple sine waves with discrete frequencies—frequencies that are integral multiples of the fundamental frequency ($1f$, $2f$, $3f$, and so on). A nonperiodic composite signal can be decomposed into a combination of an infinite number of simple sine waves with continuous frequencies, frequencies that have real values.

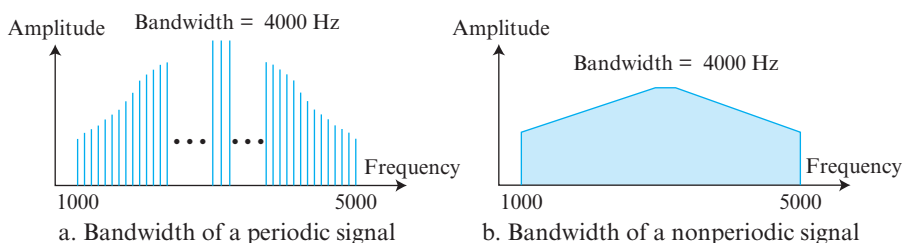
Bandwidth

The range of frequencies contained in a composite signal is its **bandwidth**. The bandwidth is normally a difference between two numbers. For example, if a composite signal contains frequencies between 1000 and 5000, its bandwidth is $5000 - 1000$, or 4000.

The bandwidth of a composite signal is the difference between the highest and the lowest frequencies contained in that signal.

Figure 7.7 shows the concept of bandwidth. The figure depicts two composite signals, one periodic and the other nonperiodic. The bandwidth of the periodic signal contains many discrete frequencies that are integral multiples of the fundamental frequency. The bandwidth of the nonperiodic signal has the same range, but the frequencies are continuous.

Figure 7.7 The bandwidth of periodic and nonperiodic composite signals



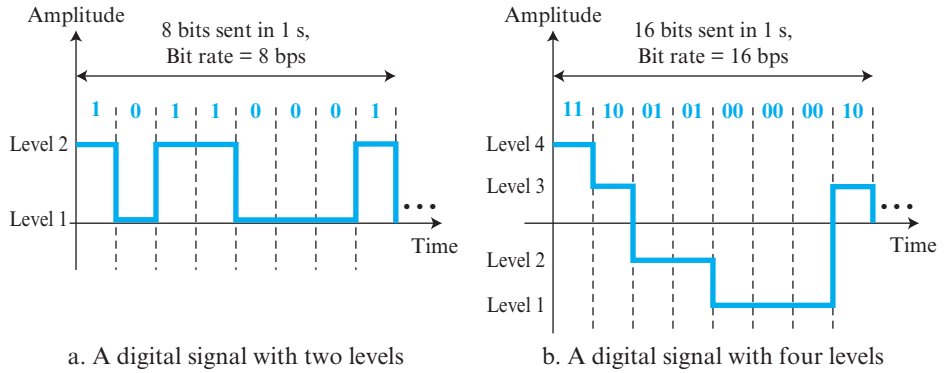
Digital Signals

In addition to being represented by an analog signal, information can also be represented by a digital signal. For example, a 1 can be encoded as a positive voltage and a 0 as zero voltage. A digital signal can have more than two levels. In this case, we can send more than 1 bit for each level. Figure 7.8 shows two signals, one with two levels and the other with four.

We send 1 bit per level in part a of the figure and 2 bits per level in part b of the figure. In general, if a signal has L levels, each level needs $\log_2 L$ bits.

Bit Rate

Most digital signals are nonperiodic, and thus period and frequency are not appropriate characteristics. Another term—*bit rate* (instead of *frequency*)—is used to

Figure 7.8 Two digital signals: one with two signal levels and the other with four signal levels

describe digital signals. The **bit rate** is the number of bits sent in 1 second, expressed in bits per second (bps).

Example 7.2

Assume we need to download text documents at the rate of 100 pages per minute. What is the required bit rate of the channel? A page is an average of 24 lines with 80 characters in each line. If we assume that one character requires 8 bits, the bit rate is

$$100 \times 24 \times 80 \times 8 = 1,536,000 \text{ bps} = 1.536 \text{ Mbps.}$$

Bit Length

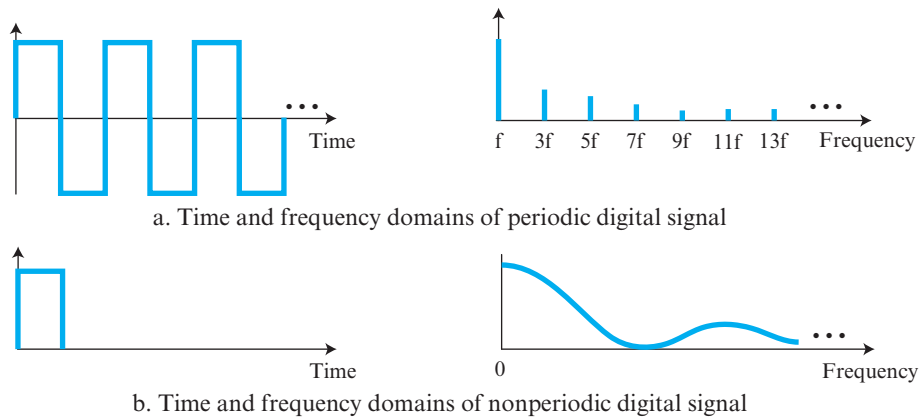
We discussed the concept of the wavelength for an analog signal: the distance one cycle occupies on the transmission medium. We can define something similar for a digital signal: the bit length. The **bit length** is the distance one bit occupies on the transmission medium.

$$\text{Bit length} = \text{propagation speed} \times \text{bit duration}$$

Digital Signal as a Composite Analog Signal

Based on Fourier analysis, a digital signal is a composite analog signal. The bandwidth is infinite, as you may have guessed. We can intuitively come up with this concept when we consider a digital signal. A digital signal, in the time domain, comprises connected vertical and horizontal line segments. A vertical line in the time domain means a frequency of infinity (sudden change in time); a horizontal line in the time domain means a frequency of zero (no change in time). Going from a frequency of zero to a frequency of infinity (and vice versa) implies all frequencies in between are part of the domain.

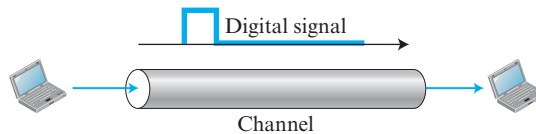
Fourier analysis can be used to decompose a digital signal. If the digital signal is periodic, which is rare in data communications, the decomposed signal has a frequency-domain representation with an infinite bandwidth and discrete frequencies. If the digital signal is nonperiodic, the decomposed signal still has an infinite bandwidth, but the frequencies are continuous. Figure 7.9 shows a periodic and a nonperiodic digital signal and their bandwidths. Note that both bandwidths are infinite, but the periodic signal has discrete frequencies while the nonperiodic signal has continuous frequencies.

Figure 7.9 The time and frequency domains of periodic and nonperiodic digital signals**Transmission of Digital Signals**

The previous discussion asserts that a digital signal, periodic or nonperiodic, is a composite analog signal with frequencies between zero and infinity. For the remainder of the discussion, let us consider the case of a nonperiodic digital signal, similar to the ones we encounter in data communications. The fundamental question is, how can we send a digital signal from point A to point B? We can transmit a digital signal by using one of two different approaches: baseband transmission or broadband transmission (using modulation).

Baseband Transmission

Baseband transmission means sending a digital signal over a channel without changing the digital signal to an analog signal. Figure 7.10 shows **baseband transmission**.

Figure 7.10 Baseband transmission

Baseband transmission requires that we have a **low-pass channel**, a channel with a bandwidth that starts from zero. This is the case if we have a dedicated medium with a bandwidth constituting only one channel. For example, the entire bandwidth of a cable connecting two computers is one single channel. As another example, we may connect several computers to a medium, but not allow more than two stations to communicate at a time. Again we have a low-pass channel, and we can use it for baseband communication.

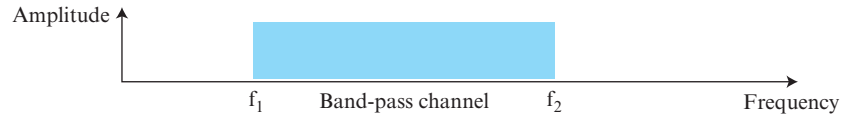
Example 7.3

An example of a dedicated channel where the entire bandwidth of the medium is used as one single channel is a LAN. Almost every wired LAN today uses a dedicated channel for two stations communicating with each other.

Broadband Transmission

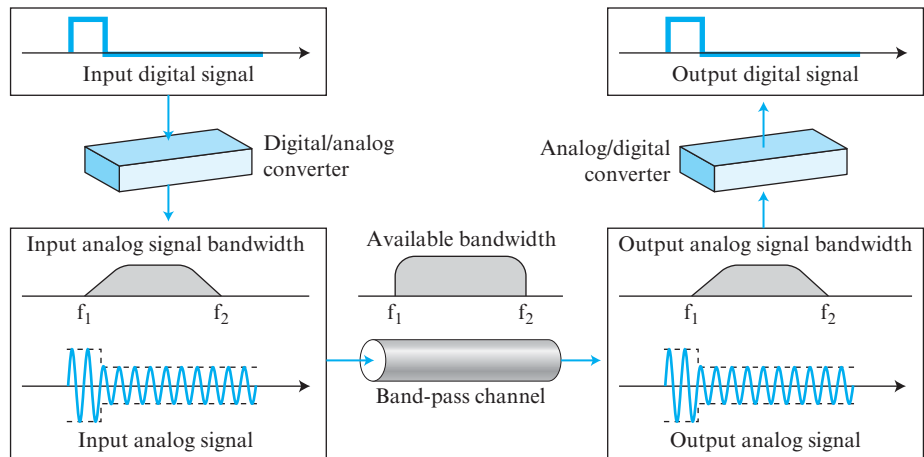
Broadband transmission or modulation means changing the digital signal to an analog signal for transmission. Modulation allows us to use a **band-pass channel**—a channel with a bandwidth that does not start from zero. This type of channel is more available than a low-pass channel. Figure 7.11 shows a bandpass channel.

Figure 7.11 Bandwidth of a band-pass channel



Note that a low-pass channel can be considered a band-pass channel with the lower frequency starting at zero. Figure 7.12 shows the modulation of a digital signal. In the figure, a digital signal is converted to a composite analog signal. We have used a single-frequency analog signal (called a *carrier*); the amplitude of the carrier has been changed to look like the digital signal. The result, however, is not a single-frequency signal; it is a composite signal, as we will see later. At the receiver, the received analog signal is converted to digital, and the result is a replica of what has been sent.

Figure 7.12 Modulation of a digital signal for transmission on a band-pass channel



Example 7.4

An example of broadband transmission using modulation is the sending of computer data through a telephone subscriber line, the line connecting a resident to the central telephone office. These lines, installed many years ago, are designed to carry voice (analog signal) with a limited bandwidth (frequencies between 0 and 4 kHz). Although this channel can be used as a low-pass channel, it is normally considered a band-pass channel. One reason is that the bandwidth is so narrow (4 kHz) that if we treat the channel as low-pass and use it for baseband transmission, the maximum bit rate can be only 8 kbps (explained later). The solution is to consider the channel a

band-pass channel, convert the digital signal from the computer to an analog signal, and send the analog signal. We can install two converters to change the digital signal to analog and vice versa at the receiving end. The converter, in this case, is called a *modem* (modulator/demodulator), which we discussed in Chapter 5.

Example 7.5

A second example is the digital cellular telephone. For better reception, digital cellular phones digitize analog voice. Although the bandwidth allocated to a company providing digital cellular phone service is very wide, we still cannot send the digitized signal without conversion. The reason is that we have only a band-pass channel available between caller and callee. For example, if the available bandwidth is W and we allow 1000 couples to talk simultaneously, this means the available channel is $W/1000$, just part of the entire bandwidth. We need to convert the digitized voice to a composite analog signal before transmission.

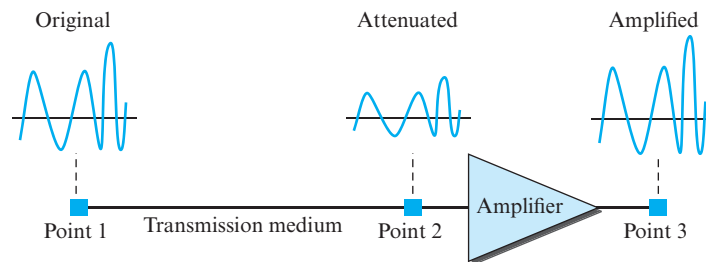
7.1.2 Transmission Impairment

Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment. This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium. What is sent is not what is received. Three causes of impairment are attenuation, distortion, and noise.

Attenuation

Attenuation means a loss of energy. When a signal, simple or composite, travels through a medium, it loses some of its energy in overcoming the resistance of the medium. That is why a wire carrying electric signals gets warm, if not hot, after a while. Some of the electrical energy in the signal is converted to heat. To compensate for this loss, amplifiers are used to amplify the signal. Figure 7.13 shows the effect of attenuation and amplification.

Figure 7.13 Attenuation and amplification



To show that a signal has lost or gained strength, engineers use the unit of the decibel. The **decibel (dB)** measures the relative strengths of two signals or one signal at two different points. Note that the decibel is negative if a signal is attenuated and positive if a signal is amplified.

$$\text{dB} = 10 \log_{10} (P_2/P_1)$$

Variables P_1 and P_2 are the powers of a signal at points 1 and 2, respectively.

Example 7.6

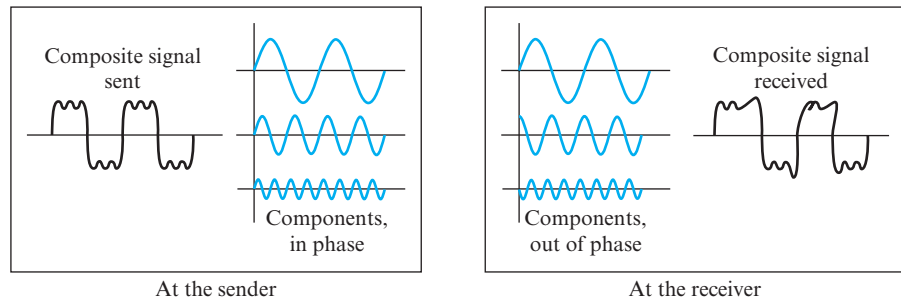
Suppose a signal travels through a transmission medium and its power is reduced to one-half. This means that $P_2 = 0.5 P_1$. In this case, the attenuation (loss of power) can be calculated as

$$10 \log_{10} P_2/P_1 = 10 \log_{10} (0.5 P_1)/P_1 = 10 \log_{10} 0.5 = 10 \times (-0.3) = -3 \text{ dB}.$$

A loss of 3 dB (−3 dB) is equivalent to losing one-half the power.

Distortion

Distortion means that the signal changes its form or shape. Distortion can occur in a composite signal made up of different frequencies. Each signal component has its own propagation speed through a medium and, therefore, its own delay in arriving at the final destination. Differences in delay may create a difference in phase if the delay is not exactly the same as the period duration. In other words, signal components at the receiver have phases different from what they had at the sender. The shape of the composite signal is therefore not the same. Figure 7.14 shows the effect of distortion on a composite signal.

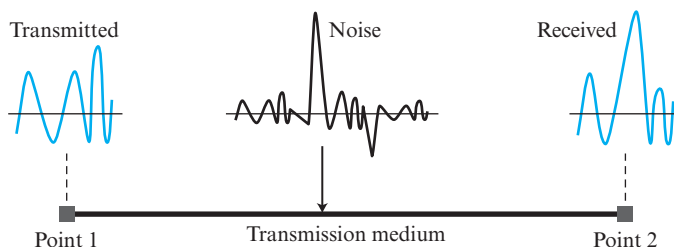
Figure 7.14 Distortion**Noise**

Noise is another cause of impairment. Several types of noise, such as thermal noise, induced noise, crosstalk, and impulse noise, may corrupt the signal. Thermal noise is the random motion of electrons in a wire, which creates an extra signal not originally sent by the transmitter. Induced noise comes from sources such as motors and appliances. These devices act as a sending antenna, and the transmission medium acts as the receiving antenna. **Crosstalk** is the effect of one wire on the other. One wire acts as a sending antenna and the other as the receiving antenna. Impulse noise is a spike (a signal with high energy and very short duration) that comes from power lines, lightning, and so on. Figure 7.15 shows the effect of noise on a signal. In Chapter 5, we discussed how to detect and control these types of errors.

Signal-to-Noise Ratio (SNR)

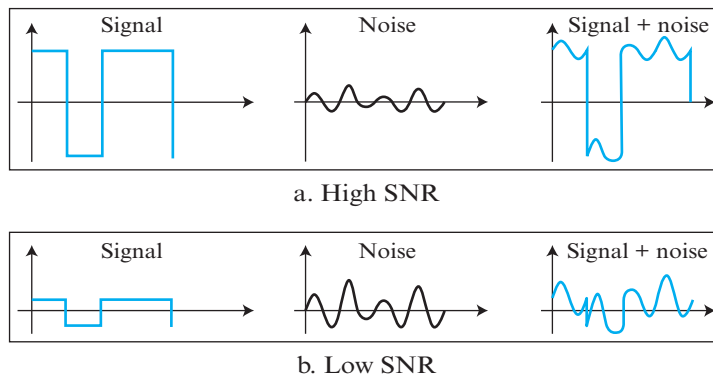
As we will see later, to find the theoretical bit-rate limit, we need to know the ratio of the signal power to the noise power. The **signal-to-noise ratio** is defined as

$$\text{SNR} = (\text{average signal power}) / (\text{average noise power}).$$

Figure 7.15 Noise

We need to consider the average signal power and the average noise power because these may change with time. Figure 7.16 shows the idea of SNR. A high SNR means the signal is less corrupted by noise; a low SNR means the signal is more corrupted by noise. Because SNR is the ratio of two powers, it is often described in decibel units, SNR_{dB} , defined below.

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR}$$

Figure 7.16 Two cases of SNR: a high SNR and a low SNR

7.1.3 Data Rate Limits

A very important consideration in data communications is how fast we can send data, in bits per second, over a channel. Data rate depends on three factors:

1. The bandwidth available
2. The level of the signals we use
3. The quality of the channel (the level of noise)

Two theoretical formulas were developed to calculate the data rate: one by Nyquist for a noiseless channel, another by Shannon for a noisy channel.

Noiseless Channel: Nyquist Bit Rate

For a noiseless channel, the **Nyquist bit rate** formula defines the theoretical maximum bit rate.

$$\text{BitRate} = 2 \times B \times \log_2 L$$

In this formula, B is the bandwidth of the channel, L is the number of signal levels used to represent data, and BitRate is the number of bits per second.

According to the formula, we might think that, given a specific bandwidth, we can have any bit rate we want by increasing the number of signal levels. Although the idea is theoretically correct, practically there is a limit. When we increase the number of signal levels, we impose a burden on the receiver. If the number of levels in a signal is just 2, the receiver can easily distinguish between a 0 and a 1. If the level of a signal is 64, the receiver must be very sophisticated to distinguish between 64 different levels. In other words, increasing the levels of a signal reduces the reliability of the system.

Example 7.7

We need to send 265 kbps over a noiseless (ideal) channel with a bandwidth of 20 kHz. How many signal levels do we need? We can use the Nyquist formula as shown:

$$265,000 = 2 \times 20,000 \times \log_2 L \rightarrow \log_2 L = 6.625 \quad L = 2^{6.625} = 98.7 \text{ levels.}$$

Since this result is not a power of 2, we need to either increase the number of levels or reduce the bit rate. If we have 128 levels, the bit rate is 280 kbps. If we have 64 levels, the bit rate is 240 kbps.

Noisy Channel: Shannon Capacity

In reality, we cannot have a noiseless channel; the channel is always noisy. In 1944, Claude Shannon introduced a formula, called the **Shannon capacity**, to determine the theoretical highest data rate for a noisy channel below.

$$C = B \log_2 (1 + \text{SNR})$$

In this formula, B is the bandwidth of the channel, SNR is the signal-to-noise ratio, and C is the capacity of the channel in bits per second. Note that in the Shannon formula there is no indication of the signal level, which means that no matter how many levels we have, we cannot achieve a data rate higher than the capacity of the channel. In other words, the formula defines a characteristic of the channel, not the method of transmission. The capacity defines the upper boundary of the channel bit rate.

Example 7.8

Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. For this channel the capacity C is calculated as shown below.

$$C = B \log_2 (1 + \text{SNR}) = B \log_2 (1 + 0) = B \log_2 1 = B \times 0 = 0$$

This means that the capacity of this channel is zero regardless of the bandwidth. In other words, the data is so corrupted in this channel that it is useless when received.

Example 7.9

We can calculate the theoretical highest bit rate of a regular telephone line. A telephone line normally has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal-to-noise ratio is usually 3162. For this channel the capacity is calculated as shown below.

$$C = B \log_2 (1 + \text{SNR}) = 3000 \log_2 (1 + 3162) = 34,881 \text{ bps}$$

This means that the highest bit rate for a telephone line is 34.881 kbps. If we want to send data faster than this, we can either increase the bandwidth of the line or improve the signal-to-noise ratio.

Using Both Limits

In practice, we need to use both methods to find the limits and signal levels. Let us show this with an example.

Example 7.10

We have a channel with a 1-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level?

Solution

First, we use the Shannon formula to find the upper limit.

$$C = B \log_2 (1 + \text{SNR}) = 10^6 \log_2 (1 + 63) = 10^6 \log_2 64 = 6 \text{ Mbps}$$

The Shannon formula gives us 6 Mbps, the upper limit. For better performance we choose something lower, 4 Mbps, for example. Then we use the Nyquist formula to find the number of signal levels.

$$4 \text{ Mbps} = 2 \times 1 \text{ MHz} \times \log_2 L \rightarrow \log_2 L = 2 \rightarrow L = 4$$

**The Shannon capacity gives us the upper limit;
the Nyquist formula tells us how many signal levels we need.**

7.1.4 Performance

Up to now, we have discussed the tools of transmitting data (signals) over a network and how the data behave. One important issue in networking is the performance of the network—how good is it? We discuss quality of service, an overall measurement of network performance, in detail in Chapter 8.

Bandwidth

One characteristic that measures network performance is bandwidth. However, the term can be used in two different contexts with two different measuring values: bandwidth in hertz and bandwidth in bits per second.

Bandwidth in Hertz

We have discussed this concept. Bandwidth in hertz is the range of frequencies contained in a composite signal or the range of frequencies a channel can pass. For example, we can say the bandwidth of a subscriber telephone line is 4 kHz.

Bandwidth in Bits per Seconds

The term *bandwidth* can also refer to the number of bits per second that a channel, a link, or even a network can transmit. For example, one can say the bandwidth of a Fast Ethernet network (or the links in this network) is 100 Mbps. This means that this network can send 100 Mbps.

Relationship

There is an explicit relationship between the bandwidth in hertz and bandwidth in bits per second. Basically, an increase in bandwidth in hertz means an increase in bandwidth in bits per second. The relationship depends on whether we use baseband transmission or broadband transmission.

Example 7.11

The bandwidth of a subscriber line is 4 kHz for voice or data. The bandwidth of this line for data transmission can be up to 56 kbps, using a sophisticated modem to change the digital signal to analog. If the telephone company improves the quality of the line and increases the bandwidth to 8 kHz, we can send 112 kbps.

Throughput

The **throughput** is a measure of how fast we can actually send data through a network. Although, at first glance, bandwidth in bits per second and throughput seem the same, they are different. A link may have a bandwidth of B bps, but we can send only T bps through this link, with T always less than B . In other words, the bandwidth is a potential measurement of a link; the throughput is an actual measurement of how fast we can send data. For example, we may have a link with a bandwidth of 1 Mbps, but the devices connected to the end of the link may handle only 200 kbps. This means that we cannot send more than 200 kbps through this link.

Imagine a highway designed to transmit 1000 cars per minute from one point to another. However, if there is congestion on the road, this figure may be reduced to 100 cars per minute. The bandwidth is 1000 cars per minute; the throughput is 100 cars per minute.

Latency (Delay)

The latency or delay defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source. We discussed the delay for a packet in Chapter 4. We said that we normally have four types of delay: propagation delay, transmission delay, queuing delay, and processing delay. The latency or total delay is

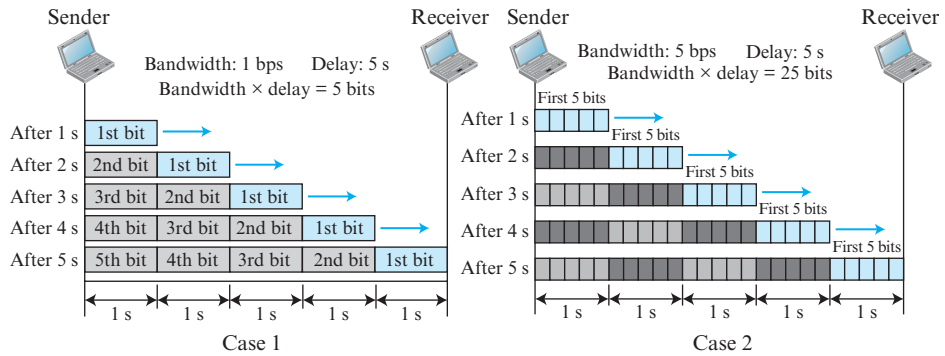
$$\text{Latency} = \text{propagation delay} + \text{transmission delay} + \text{queuing delay} + \text{processing delay}.$$

Bandwidth-Delay Product

Bandwidth and delay are two performance metrics of a link. However, what is very important in data communications is the product of the two, the bandwidth-delay product. Let us elaborate on this issue, using two hypothetical cases as examples (Figure 7.17).

- **Case 1.** Let us assume that we have a link with a bandwidth of 1 bps (unrealistic, but good for demonstration purposes). We also assume that the delay of the link is 5 seconds (also unrealistic). We want to see what the bandwidth-delay product means in this case. Looking at the figure, we can see that the bandwidth-delay product (1×5) is the maximum number of bits that can fill the link. There can be no more than 5 bits at any time on the link.
- **Case 2.** Now assume we have a bandwidth of 5 bps with a delay of 5 seconds. Figure 7.17 shows that there can be a maximum of $5 \times 5 = 25$ bits on the line. The reason is that, at each second, there are 5 bits on the line; the duration of each bit is 0.20 s.

Figure 7.17 Filling the link with bits for cases 1 and 2



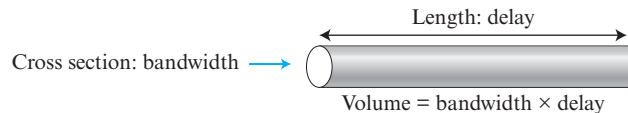
The above two cases show that the product of bandwidth and delay is the number of bits that can fill the link. This measurement is important if we need to send data in bursts and wait for the acknowledgment of each burst before sending the next one. To use the maximum capability of the link, we need to make the size of our burst 2 times the product of bandwidth and delay; we need to fill up the full-duplex channel (two directions). The sender should send a burst of data of $(2 \times \text{bandwidth} \times \text{delay})$ bits. The sender then waits for receiver acknowledgment for part of the burst before sending another burst. The amount $2 \times \text{bandwidth} \times \text{delay}$ is the number of bits that can be in transition at any time.

The bandwidth-delay product defines the number of bits that can fill the link.

Example 7.12

We can think about the link between two points as a pipe. The cross section of the pipe represents the bandwidth, and the length of the pipe represents the delay. We can say the volume of the pipe defines the bandwidth-delay product, as shown in Figure 7.18.

Figure 7.18 Concept of bandwidth-delay product



Jitter

Another performance issue that is related to delay is **jitter**. We can roughly say that jitter is a problem if different packets of data encounter different delays and the application using the data at the receiver site is time-sensitive (audio and video data, for example). If the delay for the first packet is 20 ms, for the second is 45 ms, and for the third is 40 ms, then the real-time application that uses the packets endures jitter. We discuss jitter in greater detail in Chapter 8.

7.2 DIGITAL TRANSMISSION

A computer network is designed to send information from one point to another. This information needs to be converted to either a digital signal or an analog signal for transmission. In this section, we discuss the first choice, conversion to digital signals; in the next section, we discuss the second choice, conversion to analog signals.

If data is digital, we need to use **digital-to-digital conversion** techniques, methods which convert digital data to digital signals. If data is analog, we need to use **analog-to-digital conversion** techniques, methods which change an analog signal to a digital signal.

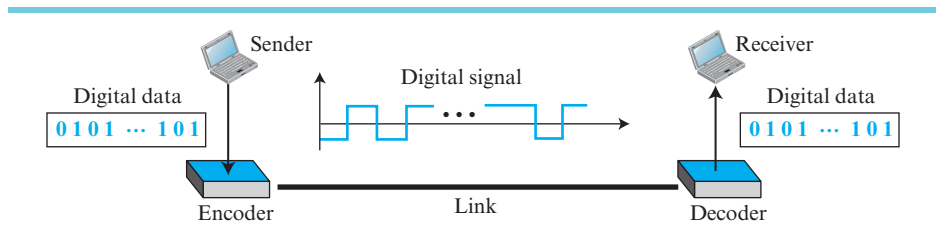
7.2.1 Digital-to-Digital Conversion

We discussed data and signals. We said that data can be either digital or analog. We also said that signals that represent data can be digital or analog. In this section, we see how we can represent digital data by using digital signals. The conversion involves three techniques: line coding, block coding, and scrambling. Line coding is always needed; block coding and scrambling may or may not be needed.

Line Coding

Line coding is the process of converting digital data to digital signals. We assume that data, in the form of text, numbers, graphical images, audio, or video, are stored in computer memory as sequences of bits. Line coding converts a sequence of bits to a digital signal. At the sender, digital data are encoded into a digital signal; at the receiver, the digital data are recreated by decoding the digital signal. Figure 7.19 shows the process.

Figure 7.19 Line coding and decoding



We can divide common line coding schemes into three categories, polar, bipolar, and multilevel, although the literature uses more categories.

Before discussing each category, let us define some terms used in line coding. We use N to show the bit rate (number of bits sent per second). We use r to show the number of bits carried by each change in the signal level. We also define S as the baud, the number of changes in the signal level per second. We are normally interested in S_{ave} , the average number of signal changes that can be calculated as shown below.

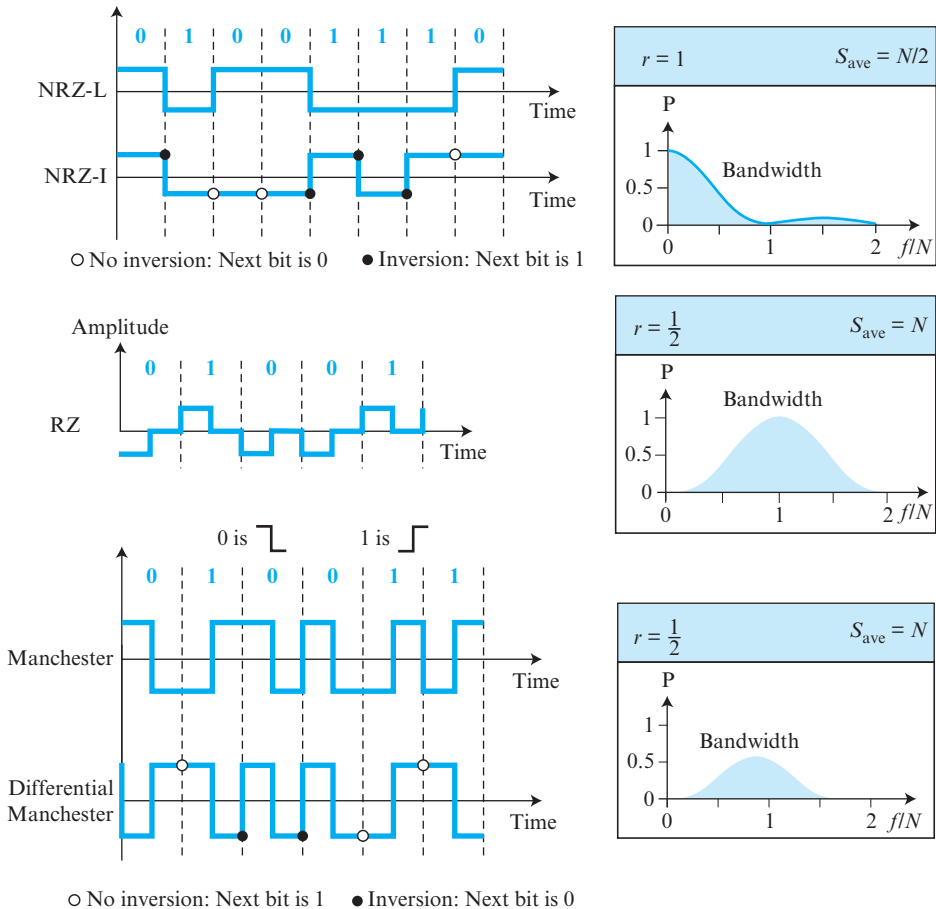
$$S_{\text{ave}} = c \times N \times (1/r)$$

In the above formula, c is a factor related to how the average is calculated. If the signal level changes when the bit changes (from 0 to 1 or from 1 to 0), the value of c is 0 if the data is made of all 1's or all 0's. It is normally 1/2 for most cases.

Polar Schemes

In polar schemes, the voltage level oscillates between a positive value and a negative value although it may remain at zero level between the two values. Figure 7.20 shows several schemes in this category. The figure also shows the bandwidth (power in the signal versus frequency), but the frequency is normalized (f/N).

Figure 7.20 Polar schemes



In **non-return-to-zero (NRZ)** schemes, we use two levels of voltage amplitude that do not stay at zero level in between. We can have two versions of polar NRZ: NRZ-L and NRZ-I. In NRZ-L (NRZ-Level), the level of the voltage determines the value of the bit. In NRZ-I (NRZ-Invert), the change or lack of change in the level of the voltage determines the value of the bit. If there is no change, the bit is 0; if there is a change, the bit is 1. If there is a long sequence of 0s or 1s, the average signal power becomes skewed. However, the skewing is twice as bad in NRZ-L. The receiver might have difficulty discerning the bit value. In NRZ-I this problem occurs only for a long sequence of 0s.

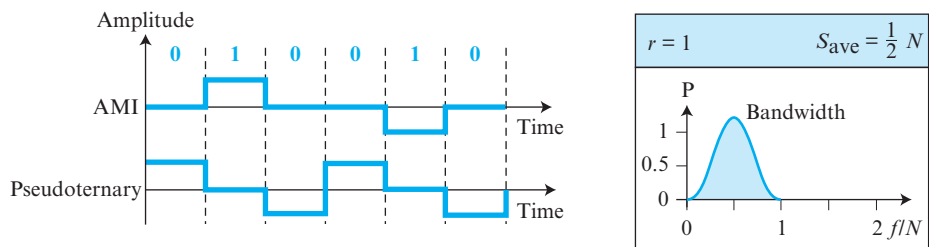
The main problem with NRZ encoding occurs when the sender and receiver clocks are not synchronized. The receiver does not know when one bit has ended and the next bit is starting. One solution is the **return-to-zero (RZ)** scheme, which uses three values: positive, negative, and zero. In RZ, the signal changes not between bits but during the bit. The main disadvantage of RZ encoding is that it requires two signal changes to encode a bit and therefore occupies greater bandwidth.

The idea of RZ (transition at the middle of the bit) and the idea of NRZ-L are combined into the **Manchester** scheme. In Manchester encoding, the duration of the bit is divided into two halves. The voltage remains at one level during the first half and moves to the other level in the second half. The transition at the middle of the bit provides synchronization. **Differential Manchester**, on the other hand, combines the ideas of RZ and NRZ-I. There is always a transition at the middle of the bit, but the bit values are determined at the beginning of the bit. If the next bit is 0, there is a transition; if the next bit is 1, there is none.

Bipolar Schemes

In **bipolar** encoding (sometimes called *multilevel binary*) there are three voltage levels: positive, negative, and zero. The voltage level for one data element is at zero, while the voltage level for the other element alternates between positive and negative. Figure 7.21 shows two variations of bipolar encoding: AMI and pseudoternary. A common bipolar encoding scheme is called bipolar **alternate mark inversion (AMI)**. In the term *alternate mark inversion*, the word *mark* comes from telegraphy and means 1. So AMI means alternate 1 inversion. A neutral zero voltage represents binary 0. Binary 1s are represented by alternating positive and negative voltages. In a variation of AMI encoding, called **pseudoternary**, the 1 bit is encoded as a zero voltage and the 0 bit is encoded as alternating positive and negative voltages.

Figure 7.21 Bipolar schemes: AMI and pseudoternary

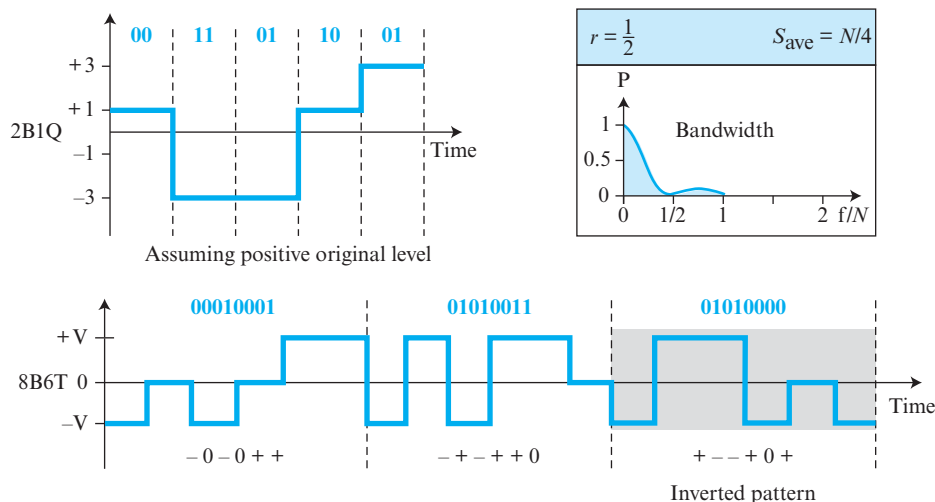


Multilevel Schemes

The desire to increase the data speed or decrease the required bandwidth has resulted in the creation of many schemes. The goal is to increase the number of bits per baud by encoding a pattern of m data elements into a pattern of n signal elements. We only have two types of data elements (0s and 1s), which means that a group of m data elements can produce a combination of 2^m data patterns. The code designers have classified these types of coding as $mBnL$, where m is the length of the binary pattern, B means binary data, n is the length of the signal pattern, and L is the number of levels in the signaling. A letter is often used in place of L : For example, $L = 2$ is B (binary), $L = 3$ is T (ternary), and $L = 4$ is Q (quaternary). Note that the first two letters define

the data pattern, and the second two define the signal pattern. We show only two of these schemes in Figure 7.22, but they are several of them.

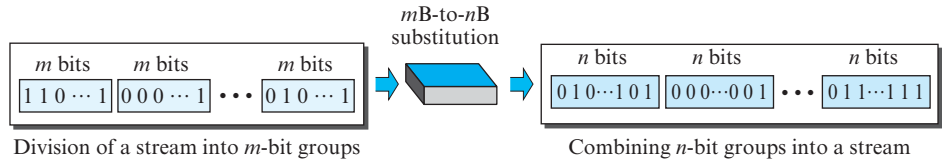
Figure 7.22 Multilevel: 2B1Q and 8B6T



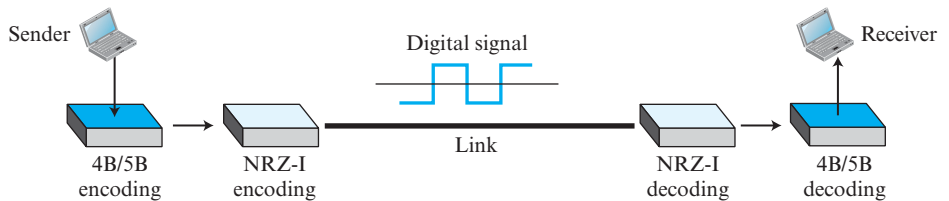
The **two binary, one quaternary (2B1Q)** scheme uses data patterns of size 2 and encodes the 2-bit patterns as one signal element belonging to a four-level signal. Note that in 2B1Q, encoding is +1, +3, -1, and -3 for the patterns 00, 01, 10, and 11. However, if the previous level is negative, the sign of the signal is inverted (-1, -3, +1, and +3). This is the case for the pattern 01 in the figure that has changed from +3 to -3 because of the previous signal level. A very interesting scheme is **eight binary, six ternary (8B6T)**. This code is used with 100BASE-4T cable, as we saw in Chapter 5. The idea is to encode a pattern of 8 bits as a pattern of 6 signal elements, where the signal has three levels (ternary). In this type of scheme, we can have $2^8 = 256$ different data patterns and $3^6 = 729$ different signal patterns. There are $729 - 256 = 473$ redundant signal elements that provide synchronization and error detection. Note that the third pattern is inverted to avoid a DC component, a non-zero voltage average.

Block Coding

We need redundancy to ensure synchronization and to provide some kind of inherent error detecting. Block coding can give us this redundancy and improve the performance of line coding. In general, **block coding** changes a block of m bits into a block of n bits, where n is larger than m . Block coding is referred to as an mB/nB encoding technique. The slash in block encoding (for example, 4B/5B) distinguishes block encoding from multilevel encoding (for example, 8B6T), which is written without a slash. Block coding normally involves three steps: division, substitution, and combination. In the division step, a sequence of bits is divided into groups of m bits. For example, in 4B/5B encoding, the original bit sequence is divided into 4-bit groups. The heart of block coding is the substitution step. In this step, we substitute an m -bit group for an n -bit group. For example, in 4B/5B encoding we substitute a 4-bit code for a 5-bit group. Finally, the n -bit groups are combined to form a stream. The new stream has more bits than the original bits. Figure 7.23 shows the procedure.

Figure 7.23 Block coding concept**4B/5B Coding**

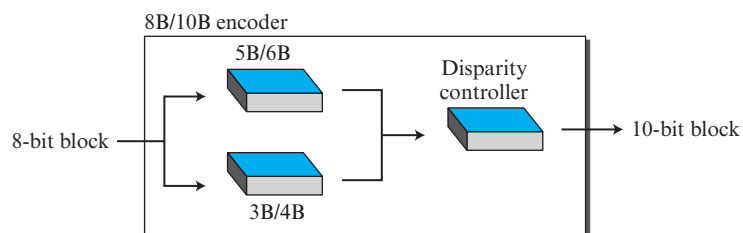
The **four binary/five binary (4B/5B)** coding scheme was designed to be used in combination with line coding schemes such as NRZ-I, in which a long sequence of 0s can make the receiver clock lose synchronization. One solution is to change the bit stream, prior to encoding with NRZ-I, so that it does not have a long stream of 0s. The 4B/5B scheme achieves this goal. The block-coded stream does not have more than three consecutive 0s. At the receiver, the NRZ-I encoded digital signal is first decoded into a stream of bits and then decoded to remove the redundancy. Figure 7.24 shows the idea.

Figure 7.24 Using block coding 4B/5B with NRZ-I line coding scheme

In 4B/5B, the 5-bit output that replaces the 4-bit input has no more than one leading zero (left bit) and no more than two trailing zeros (right bits). So when different groups are combined to make a new sequence, there are never more than three consecutive 0s.

8B/10B Coding

Eight binary/ten binary (8B/10B) encoding is similar to 4B/5B encoding except that a group of 8 bits of data is now replaced by a 10-bit code. It provides greater error detection capability than 4B/5B. The 8B/10B block coding is actually a combination of 5B/6B and 3B/4B encoding, as shown in Figure 7.25.

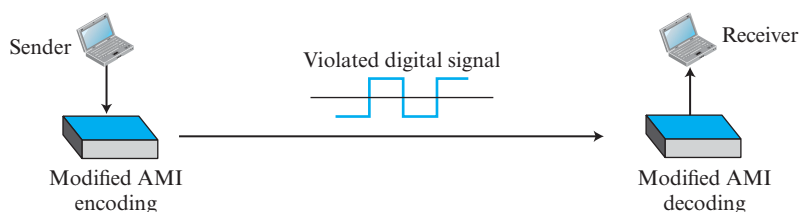
Figure 7.25 8B/10B block encoding

The five most significant bits of a 10-bit block are fed into the 5B/6B encoder; the three least significant bits are fed into a 3B/4B encoder. The split is done to simplify the mapping table. To prevent a long run of consecutive 0s or 1s, the code uses a disparity controller which keeps track of excess 0s over 1s (or 1s over 0s). If the bits in the current block create a disparity that contributes to the previous disparity (either direction), then each bit in the code is complemented (a 0 is changed to a 1 and a 1 is changed to a 0). The coding has $2^{10} - 2^8 = 768$ redundant groups that can be used for disparity checking and error detection. In general, the technique is superior to 4B/5B because of better built-in error-checking capability and better synchronization.

Scrambling

Biphase schemes that are suitable for dedicated links between stations in a LAN are not suitable for long-distance communication because of their wide bandwidth requirement. The combination of block coding and NRZ line coding is not suitable for long-distance encoding either, because of the DC component (a signal with frequency of 0). Bipolar AMI encoding, on the other hand, has a narrow bandwidth and does not create a DC component. However, a long sequence of 0s upsets the synchronization. If we can find a way to avoid a long sequence of 0s in the original stream, we can use bipolar AMI for long distances. We are looking for a technique that does not increase the number of bits and does provide synchronization. In other words, we are looking for a solution that substitutes long zero-level pulses with a combination of other levels to provide synchronization. One solution is called **scrambling**. We modify part of the AMI rule to include scrambling, as shown in Figure 7.26. Note that scrambling, as opposed to block coding, is done at the same time as encoding. The system needs to insert the required pulses based on the defined scrambling rules. Two common scrambling techniques are B8ZS and HDB3.

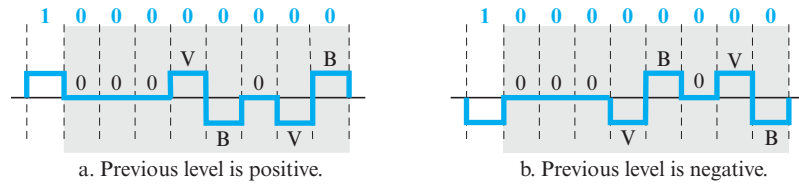
Figure 7.26 AMI used with scrambling



B8ZS Coding

Bipolar with 8-zero substitution (B8ZS) is commonly used in North America. In this technique, eight consecutive zero-level voltages are replaced by the sequence **000VB0VB**. The V in the sequence denotes *violation*; this is a nonzero voltage that breaks an AMI rule of encoding (opposite polarity from the previous). The B in the sequence denotes *bipolar*, which means a nonzero level voltage in accordance with the AMI rule. There are two cases, as shown in Figure 7.27.

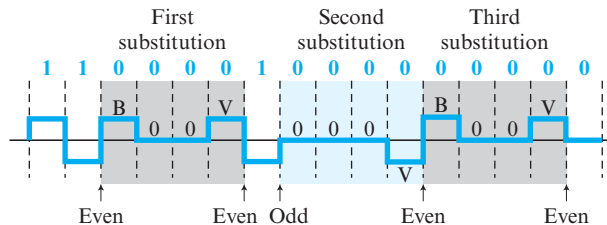
Note that the scrambling in this case does not change the bit rate. Also, the technique balances the positive and negative voltage levels (two positives and two negatives), which means that the DC balance is maintained. Note that the substitution may change the polarity of a 1 because, after the substitution, AMI needs to follow its rules.

Figure 7.27 Two cases of B8ZS scrambling technique

One more point is worth mentioning. The letter V (violation) or B (bipolar) here is relative. The V means the same polarity as the polarity of the previous nonzero pulse; B means the polarity opposite to the polarity of the previous nonzero pulse.

HDB3 Coding

High-density bipolar 3-zero (HDB3) is commonly used outside of North America. In this technique, which is more conservative than B8ZS, four consecutive zero-level voltages are replaced with a sequence of **000V** or **B00V**. The reason for two different substitutions is to maintain the even number of nonzero pulses after each substitution. The two rules can be stated as follows: If the number of nonzero pulses after the last substitution is odd, the substitution pattern will be **000V**, which makes the total number of nonzero pulses even. If the number of nonzero pulses after the last substitution is even, the substitution pattern will be **B00V**, which also makes the total number of nonzero pulses even. Figure 7.28 shows an example.

Figure 7.28 Different situations in HDB3 scrambling technique

There are several points we need to mention here. First, before the first substitution, the number of nonzero pulses is even, so the first substitution is B00V. After this substitution, the polarity of the 1 bit is changed because the AMI scheme, after each substitution, must follow its own rule. After this bit, we need another substitution, which is 000V because we have only one nonzero pulse (odd) after the last substitution. The third substitution is B00V because there are no nonzero pulses after the second substitution (even).

7.2.2 Analog-to-Digital Conversion

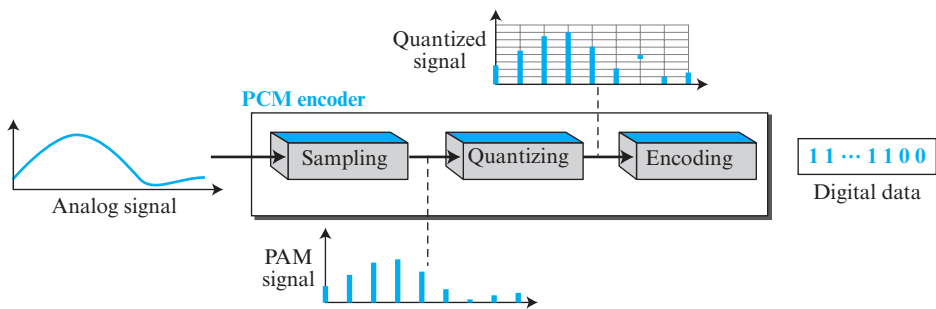
The techniques described in Section 7.2.1 convert digital data to digital signals. Sometimes, however, we have an analog signal such as one created by a microphone or camera. The tendency today is to change an analog signal to digital data because the digital signal is less susceptible to noise. In this section we describe two techniques, pulse code modulation and delta modulation. After the digital data are created (digitization),

we can use one of the techniques described in the previous section to convert the digital data to a digital signal.

Pulse Code Modulation (PCM)

The most common technique to change an analog signal to digital data (**digitization**) is called **pulse code modulation (PCM)**. A PCM encoder has three processes, as shown in Figure 7.29.

Figure 7.29 Components of PCM encoder



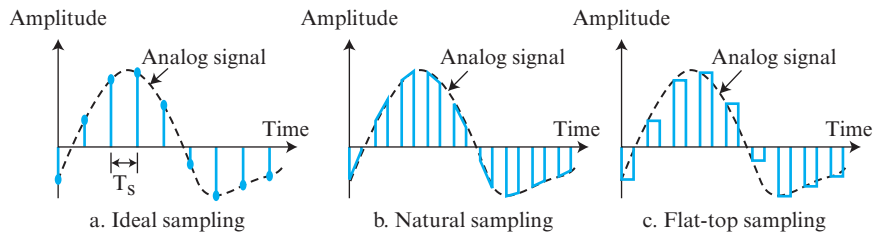
The three processes are:

1. The analog signal is sampled.
2. The sampled signal is quantized.
3. The quantized values are encoded as streams of bits.

Sampling

The first step in PCM is **sampling**. The analog signal is sampled every T_s seconds, where T_s is the sample interval or period. The inverse of the sampling interval is called the **sampling rate** or **sampling frequency** and denoted by f_s , where $f_s = 1/T_s$. There are three sampling methods—ideal, natural, and flat-top—as shown in Figure 7.30.

Figure 7.30 Three different sampling methods for PCM



In ideal sampling, pulses from the analog signal are sampled. This is an ideal sampling method and cannot be easily implemented. In natural sampling, a high-speed switch is turned on for only the small period of time when the sampling occurs. The result is a sequence of samples that retains the shape of the analog signal. The most

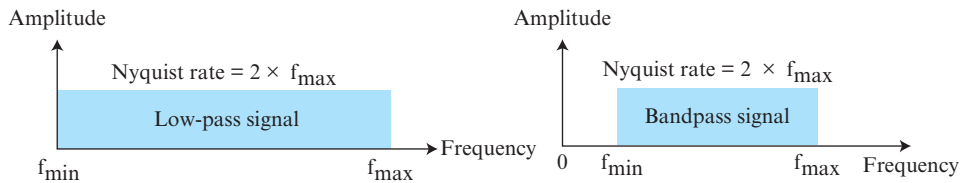
common sampling method, called **sample and hold**, however, creates flat-top samples by using a circuit.

The sampling process is sometimes referred to as **pulse amplitude modulation (PAM)**. We need to remember, however, that the result is still an analog signal with nonintegral values.

Sampling Rate One important consideration is the sampling rate or frequency. What are the restrictions on T_s ? This question was elegantly answered by Nyquist. According to the **Nyquist theorem**, to reproduce the original analog signal, one necessary condition is that the **sampling rate** be at least twice the highest frequency in the original signal.

We need to elaborate on the theorem at this point. First, we can sample a signal only if the signal is band-limited. In other words, a signal with an infinite bandwidth cannot be sampled. Second, the sampling rate must be at least two times the highest frequency, not the bandwidth. If the analog signal is low-pass, the bandwidth and the highest frequency are the same value. If the analog signal is bandpass, the bandwidth value is lower than the value of the maximum frequency. Figure 7.31 shows the value of the sampling rate for two types of signals.

Figure 7.31 Nyquist sampling rate for low-pass and bandpass signals



Quantization

The result of sampling is a series of pulses with amplitude values between the maximum and minimum amplitudes of the signal. The set of amplitudes can be infinite with nonintegral values between the two limits. These values cannot be used in the encoding process. The following are the steps in quantization:

1. We assume that the original analog signal has values between $V_{\min}$ and $V_{\max}$.
2. We divide the range into L zones, each of height Δ (delta).

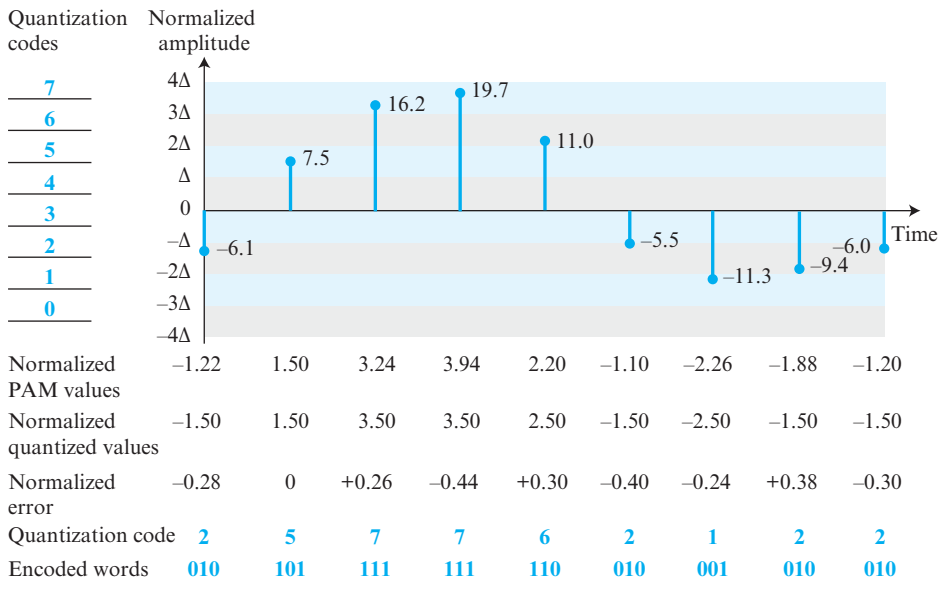
$$\Delta = (V_{\max} - V_{\min}) / L$$

3. We assign quantized values of 0 to $L - 1$ to the midpoint of each zone.
4. We approximate the value of the sample amplitude to the quantized values.

As a simple example, assume that we have a sampled signal and the sample amplitudes are between -20 and $+20$ V. We decide to have eight levels ($L = 8$). This means that the value for $\Delta = 5$ V. Figure 7.32 shows this example.

We have shown only nine samples using ideal sampling (for simplicity). The value at the top of each sample in the graph shows the actual amplitude. In the chart, the first row is the normalized value for each sample (actual amplitude/ Δ). The quantization process selects the quantization value from the middle of each zone. This means that the normalized quantized values (second row) are different from the normalized

Figure 7.32 Quantization and encoding of a sampled signal



amplitudes. The difference is called the *normalized error* (third row). The fourth row is the quantization code for each sample based on the quantization levels at the left of the graph. The encoded words (fifth row) are the final products of the conversion.

Quantization Levels In the previous example, we showed eight quantization levels. The choice of L , the number of levels, depends on the range of the amplitudes of the analog signal and how accurately we need to recover the signal. If the amplitude of a signal fluctuates between two values only, we need only two levels; if the signal, like voice, has many amplitude values, we need more quantization levels. In audio digitizing, L is normally chosen to be 256; in video it is normally thousands. Choosing lower values of L increases the quantization error if there is a lot of fluctuation in the signal.

Quantization Error One important issue is the error created in the quantization process. Quantization is an approximation process. The input values to the quantizer are the real values; the output values are the approximated values. Each output value is chosen to be the middle value in the zone. If the input value is also at the middle of the zone, there is no quantization error; otherwise, there is an error. In the previous example, the normalized amplitude of the third sample is 3.24, but the normalized quantized value is 3.50. This means that there is an error of +0.26. The value of the error for any sample is less than $\Delta/2$. In other words, we have $-\Delta/2 \leq \text{error} \leq \Delta/2$.

The quantization error changes the signal-to-noise ratio of the signal, which in turn reduces the upper limit capacity according to Shannon.

It can be proven that the contribution of the *quantization error* to the SNR_{dB} of the signal depends on the number of quantization levels L , or the bits per sample n_b , as shown in the following formula.

$$\text{SNR}_{\text{dB}} = 6.02n_b + 1.76 \text{ dB}$$

Encoding

The last step in PCM is encoding. After each sample is quantized and the number of bits per sample is decided, each sample can be changed to an n_b -bit code word. In Figure 7.32 the encoded words are shown in the last row. A quantization code of 2 is encoded as 010; 5 is encoded as 101, and so on. Note that the number of bits for each sample is determined from the number of quantization levels. If the number of quantization levels is L , the number of bits is $n_b = \log_2 L$. In our example L is 8 and n_b is therefore 3. The bit rate can be found from the formula shown below.

$$\text{Bit rate} = \text{sampling rate} \times \text{number of bits per sample} = f_s \times n_b$$

Example 7.13

We want to digitize the human voice. What is the bit rate, assuming 8 bits per sample?

Solution

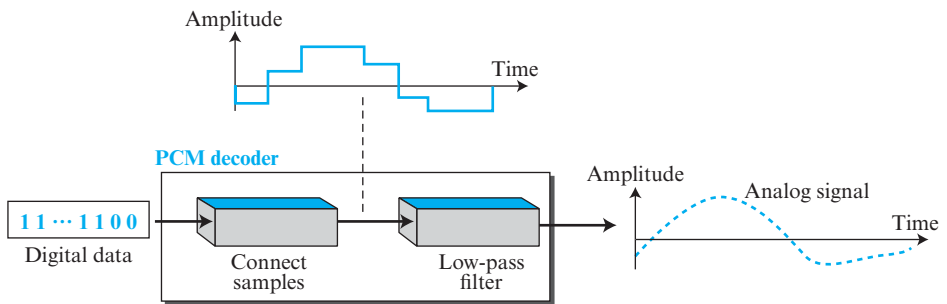
The human voice normally contains frequencies from 0 to 4000 Hz. So the sampling rate and bit rate are calculated as follows.

$$\text{Sampling rate} = 4000 \times 2 = 8000 \text{ samples/s} \rightarrow \text{Bit rate} = 8000 \times 8 = 64,000 \text{ bps} = 64 \text{ kbps}$$

Original Signal Recovery

The recovery of the original signal requires the PCM decoder. The decoder first uses circuitry to convert the code words into a pulse that holds the amplitude until the next pulse. After the staircase signal is completed, it is passed through a low-pass filter to smooth the staircase signal into an analog signal. The filter has the same cutoff frequency as the original signal at the sender. If the signal has been sampled at (or above) the Nyquist sampling rate and if there are enough quantization levels, the original signal will be recreated. Note that the maximum and minimum values of the original signal can be achieved by using amplification. Figure 7.33 shows the simplified process.

Figure 7.33 Components of a PCM decoder



PCM Bandwidth

It can be proved that the minimum bandwidth of the digital signal is shown below.

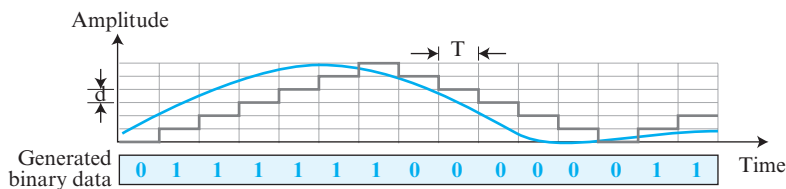
$$B_{\min} = n_b \times B_{\text{analog}}$$

This means the minimum bandwidth of the digital signal is n_b times greater than the bandwidth of the analog signal. This is the price we pay for digitization.

Delta Modulation (DM)

PCM is a very complex technique. Other techniques have been developed to reduce the complexity of PCM. The simplest is **delta modulation**. PCM finds the value of the signal amplitude for each sample; DM finds the change from the previous sample. Figure 7.34 shows the process. Note that there are no code words here; bits are sent one after another.

Figure 7.34 The process of delta modulation



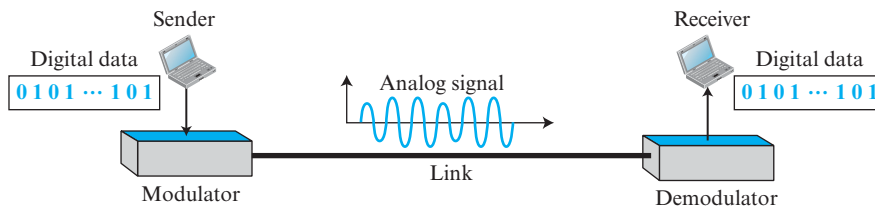
7.3 ANALOG TRANSMISSION

While digital transmission is desirable, it needs a low-pass channel; analog transmission is the only choice if we have a bandpass channel. Converting digital data to a bandpass analog signal is traditionally called *digital-to-analog conversion*. Converting a low-pass analog signal to a bandpass analog signal is traditionally called *analog-to-analog conversion*. In this section, we discuss these two types of conversions.

7.3.1 Digital-to-Analog Conversion

Digital-to-analog conversion is the process of changing one of the characteristics of an analog signal based on the information in digital data. Figure 7.35 shows the relationship between the digital information, the digital-to-analog modulating process, and the resultant analog signal.

Figure 7.35 Digital-to-analog conversion



As discussed before, a sine wave is defined by three characteristics: amplitude, frequency, and phase. When we vary any one of these characteristics, we create a different version of that wave. So, by changing one characteristic of a simple electric signal, we

can use it to represent digital data. Any of the three characteristics can be altered in this way, giving us at least three mechanisms for modulating digital data into an analog signal: **amplitude shift keying (ASK)**, **frequency shift keying (FSK)**, and **phase shift keying (PSK)**. In addition, there is a fourth (and better) mechanism that combines changing both the amplitude and phase, called **quadrature amplitude modulation (QAM)**. QAM is the most efficient of these options and is the mechanism commonly used today.

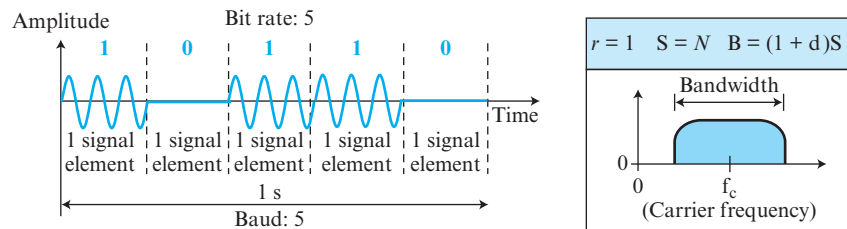
Amplitude Shift Keying

In amplitude shift keying, the amplitude of the carrier signal is varied to create signal elements. Both frequency and phase remain constant while the amplitude changes.

Binary ASK (BASK)

ASK is normally implemented using only two levels. This is referred to as *binary amplitude shift keying* or *on-off keying (OOK)*. The peak amplitude of one signal level is 0; the other is the same as the amplitude of the carrier frequency. Figure 7.36 gives a conceptual view of binary ASK. Figure 7.36 also shows the bandwidth for ASK. Although the carrier signal is only one simple sine wave, the process of modulation produces a nonperiodic composite signal. This signal, as was discussed before, has a continuous set of frequencies. As we expect, the bandwidth is proportional to the signal rate (baud). However, there is normally another factor involved, called d , which depends on the modulation and filtering process. The value of d is between 0 and 1. This means that the bandwidth can be expressed as shown, where S is the signal rate and B is the bandwidth.

Figure 7.36 Binary amplitude shift keying



The formula, $B = (1 + d)S$, shows that the required bandwidth has a minimum value of S and a maximum value of $2S$. The most important point here is the location of the bandwidth. The middle of the bandwidth is where f_c , the carrier frequency, is located. This means if we have a bandpass channel available, we can choose our f_c so that the modulated signal occupies that bandwidth. This is in fact the most important advantage of digital-to-analog conversion. We can shift the resulting bandwidth to match what is available.

Multilevel ASK

The above discussion uses only two amplitude levels. We can have multilevel ASK in which there are more than two levels. We can use 4, 8, 16, or more different amplitudes for the signal and modulate the data using 2, 3, 4, or more bits at a time. In these cases, $r=2$, $r=3$, $r=4$, and so on. Although this is not implemented with pure ASK, it is implemented with QAM (as we will see later).

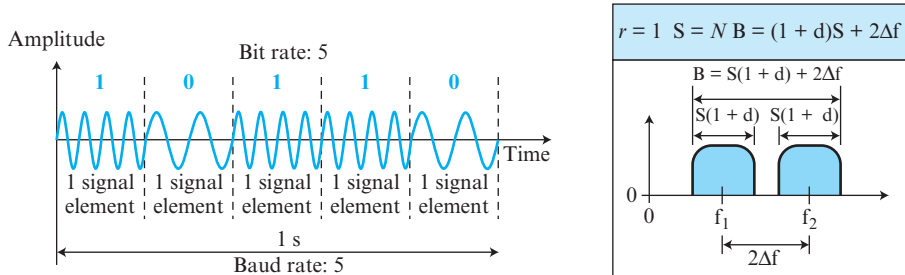
Frequency Shift Keying (FSK)

In frequency shift keying, the frequency of the carrier signal is varied to represent data. The frequency of the modulated signal is constant for the duration of one signal element, but changes for the next signal element if the data element changes. Both peak amplitude and phase remain constant for all signal elements.

Binary FSK (BFSK)

One way to think about binary FSK (or BFSK) is to consider two carrier frequencies. In Figure 7.37, we have selected two carrier frequencies, f_1 and f_2 . We use the first carrier if the data element is 0; we use the second if the data element is 1. However, note that this is an unrealistic example used only for demonstration purposes. Normally the carrier frequencies are very high, and the difference between them is very small.

Figure 7.37 Binary frequency shift keying



As Figure 7.37 shows, the middle of one bandwidth is f_1 and the middle of the other is f_2 . Both f_1 and f_2 are Δf apart from the midpoint between the two bands. The difference between the two frequencies is $2\Delta f$.

Figure 7.37 also shows the bandwidth of FSK. Again the carrier signals are only simple sine waves, but the modulation creates a nonperiodic composite signal with continuous frequencies. We can think of FSK as two ASK signals, each with its own carrier frequency (f_1 or f_2). If the difference between the two frequencies is $2\Delta f$, then the required bandwidth is $B = (1 + d) \times S + 2\Delta f$.

What should be the minimum value of $2\Delta f$? In Figure 7.37, we have chosen a value greater than $(1 + d)S$. It can be shown that the minimum value should be at least S for the proper operation of modulation and demodulation.

Multilevel FSK

Multilevel modulation (MFSK) is not uncommon with the FSK method. We can use more than two frequencies. For example, we can use four different frequencies f_1, f_2, f_3 , and f_4 to send 2 bits at a time. To send 3 bits at a time, we can use eight frequencies. And so on. However, we need to remember that the frequencies need to be $2\Delta f$ apart. For the proper operation of the modulator and demodulator, it can be shown that the minimum value of $2\Delta f$ needs to be S . We can show that the bandwidth with $d = 0$ is

$$B = (1 + d) \times S + (L - 1)2\Delta f \rightarrow B = L \times S.$$

Phase Shift Keying

In phase shift keying, the phase of the carrier is varied to represent two or more different signal elements. Both peak amplitude and frequency remain constant as the phase changes. Today, PSK is more common than ASK or FSK. However, we will see shortly that QAM, which combines ASK and PSK, is the dominant method of digital-to-analog modulation.

Binary PSK (BPSK)

The simplest PSK is binary BPSK, in which we have only two signal elements, one with a phase of 0° , and the other with a phase of 180° . Figure 7.38 gives a conceptual view of PSK. Binary PSK is as simple as binary ASK with one big advantage—it is less susceptible to noise. In ASK, the criterion for bit detection is the amplitude of the signal; in PSK, it is the phase. Noise can change the amplitude more easily than it can change the phase. PSK is superior to FSK because we do not need two carrier signals.

Figure 7.38 Binary phase shift keying

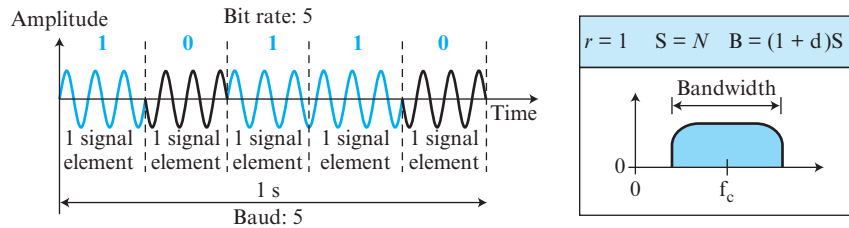


Figure 7.38 also shows the bandwidth for BPSK. The bandwidth is the same as that for binary ASK, but less than that for BFSK. No bandwidth is wasted for separating two carrier signals.

Quadrature PSK (QPSK)

The simplicity of BPSK enticed designers to use 2 bits at a time in each signal element, thereby decreasing the baud rate and eventually the required bandwidth. The scheme is called *quadrature PSK* or *QPSK* because it uses two separate BPSK modulations; one is in-phase, the other quadrature (out-of-phase by 90 degrees). The incoming bits are first passed through a serial-to-parallel conversion that sends one bit to one modulator and the next bit to the other modulator. If the duration of each bit in the incoming signal is T , the duration of each bit sent to the corresponding BPSK signal is $2T$. This means that the bit to each BPSK signal has one-half the frequency of the original signal.

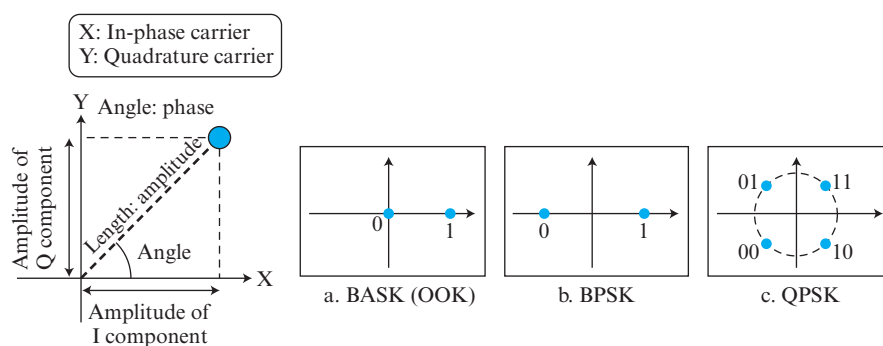
Constellation Diagram

A **constellation diagram** can help us define the amplitude and phase of a signal element, particularly when we are using two carriers (one in-phase and one quadrature). The diagram is useful when we are dealing with multilevel ASK, PSK, or QAM (see next section). In a constellation diagram, a signal element type is represented as a dot. The bit or combination of bits it can carry is often written next to it.

The diagram has two axes. The horizontal X axis is related to the in-phase carrier; the vertical Y axis is related to the quadrature carrier. For each point on the diagram, four pieces of information can be deduced. The projection of the point on the X axis

defines the peak amplitude of the in-phase component; the projection of the point on the Y axis defines the peak amplitude of the quadrature component. The length of the line (vector) that connects the point to the origin is the peak amplitude of the signal element (combination of the X and Y components); the angle the line makes with the X axis is the phase of the signal element. All the information we need can easily be found on a constellation diagram. Figure 7.39 shows the idea with three constellation diagrams, one each for BASK (OOK), BPSK, and QPSK signals.

Figure 7.39 Concept of a constellation diagram

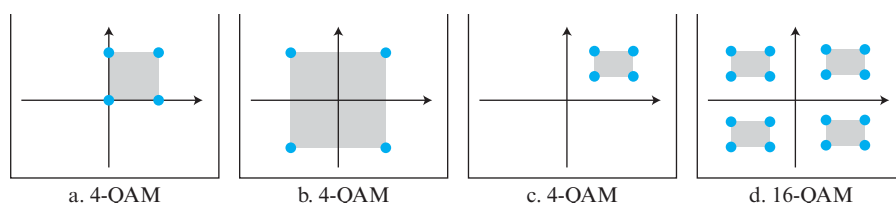


Quadrature Amplitude Modulation

PSK is limited by the ability of the equipment to distinguish small differences in phase. This factor limits its potential bit rate. So far, we have been altering only one of the three characteristics of a sine wave at a time; but what if we alter two? Why not combine ASK and PSK? The idea of using two carriers, one in-phase and the other quadrature, with different amplitude levels for each carrier, is the concept behind **quadrature amplitude modulation (QAM)**.

The possible variations of QAM are numerous. Figure 7.40 shows some of these schemes. Figure 7.40a shows the simplest 4-QAM scheme (four different signal element types) using a unipolar NRZ signal to modulate each carrier. This is the same mechanism we used for BASK (OOK). Part b shows another 4-QAM using polar NRZ, but this is exactly the same as QPSK. Part c shows another 4-QAM in which we used a signal with two positive levels to modulate each of the two carriers. Finally, Figure 7.40d shows a 16-QAM constellation of a signal with eight levels, four positive and four negative.

Figure 7.40 Constellation diagrams for some QAMs



Bandwidth for QAM

The minimum bandwidth required for QAM transmission is the same as that required for ASK and PSK transmission. QAM has the same advantages as PSK over ASK.

7.3.2 Analog-to-Analog Conversion

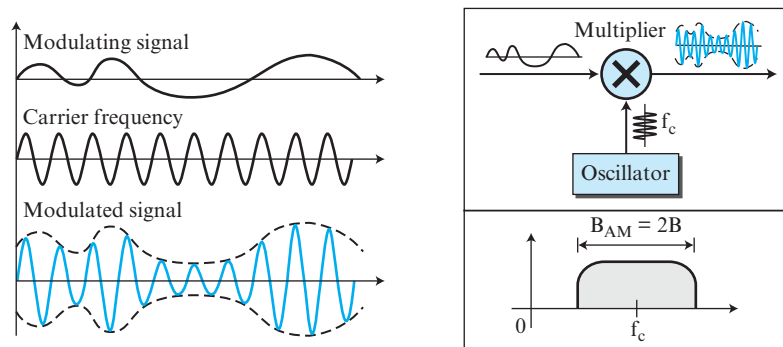
Analog-to-analog conversion, or analog modulation, is the representation of analog information by an analog signal. One may ask why we need to modulate an analog signal; it is already analog. Modulation is needed if the medium is bandpass in nature or if only a bandpass channel is available to us. An example is radio. The government assigns a narrow bandwidth to each radio station. The analog signal produced by each station is a low-pass signal, all in the same range. To be able to listen to different stations, the low-pass signals need to be shifted, each to a different range.

Analog-to-analog conversion can be accomplished in three ways: **amplitude modulation (AM)**, **frequency modulation (FM)**, and **phase modulation (PM)**. FM and PM are usually categorized together.

Amplitude Modulation

In AM transmission, the carrier signal is modulated so that its amplitude varies with the changing amplitudes of the modulating signal. The frequency and phase of the carrier remain the same; only the amplitude changes to follow variations in the information. Figure 7.41 shows how this concept works. The modulating signal is the envelope of the carrier.

Figure 7.41 Amplitude modulation

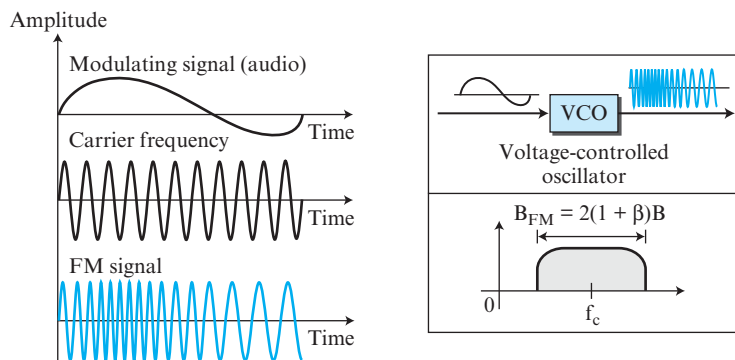


As Figure 7.41 shows, AM is normally implemented by using a simple multiplier because the amplitude of the carrier signal needs to be changed according to the amplitude of the modulating signal. Figure 7.41 also shows the bandwidth of an AM signal. The modulation creates a bandwidth that is twice the bandwidth of the modulating signal and covers a range centered on the carrier frequency. However, the signal components above and below the carrier frequency carry exactly the same information. For this reason, some implementations discard one-half of the signals and cut the bandwidth in half.

Frequency Modulation

In FM transmission, the frequency of the carrier signal is modulated to follow the changing voltage level (amplitude) of the modulating signal. The peak amplitude and phase of the carrier signal remain constant, but as the amplitude of the information signal changes, the frequency of the carrier changes correspondingly. Figure 7.42 shows the relationships of the modulating signal, the carrier signal, and the resultant FM signal.

Figure 7.42 Frequency modulation



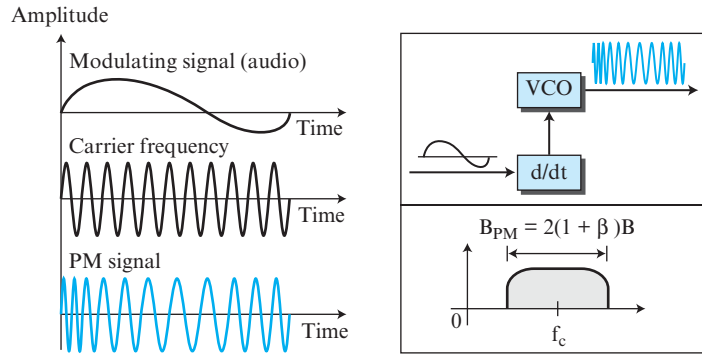
As Figure 7.42 shows, FM is normally implemented by using a voltage-controlled oscillator as with FSK. The frequency of the oscillator changes according to the input voltage, which is the amplitude of the modulating signal. Figure 7.42 also shows the bandwidth of an FM signal. The actual bandwidth is difficult to determine exactly, but it can be shown empirically as $B_{FM} = 2(1 + \beta)B$ where β is a factor that depends on modulation technique with a common value of 4.

Phase Modulation

In PM transmission, the phase of the carrier signal is modulated to follow the changing voltage level (amplitude) of the modulating signal. The peak amplitude and frequency of the carrier signal remain constant, but as the amplitude of the information signal changes, the phase of the carrier changes correspondingly. It can be proven mathematically that PM is the same as FM with one difference. In FM, the instantaneous change in the carrier frequency is proportional to the amplitude of the modulating signal; in PM the instantaneous change in the carrier frequency is proportional to the derivative of the amplitude of the modulating signal. Figure 7.43 shows the relationships of the modulating signal, the carrier signal, and the resultant PM signal.

As Figure 7.43 shows, PM is normally implemented by using a voltage-controlled oscillator along with a derivative. The frequency of the oscillator changes according to the derivative of the input voltage, which is the amplitude of the modulating signal.

Figure 7.43 also shows the bandwidth of a PM signal. The actual bandwidth is difficult to determine exactly, but it can be shown empirically that it is several times that of the analog signal. Although the formula shows the same bandwidth for FM and PM, the value of β is lower in the case of PM (around 1 for narrowband and 3 for wideband).

Figure 7.43 Phase modulation

7.4 BANDWIDTH UTILIZATION

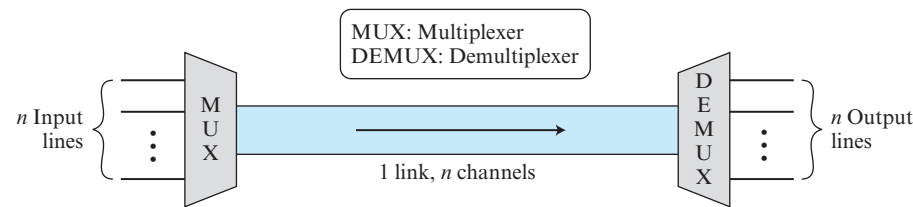
In real life, we have links with limited bandwidths. The wise use of these bandwidths has been, and will be, one of the main challenges of electronic communications. However, the meaning of *wise* may depend on the application. Sometimes we need to combine several low-bandwidth channels to make use of one channel with a larger bandwidth. Sometimes we need to expand the bandwidth of a channel to achieve goals such as privacy and anti-jamming. In this section, we explore these two broad categories of bandwidth utilization: *multiplexing* and *spreading*. In multiplexing, our goal is efficiency; we combine several channels into one. In spreading, our goals are privacy and anti-jamming; we expand the bandwidth of a channel to insert redundancy, which is necessary to achieve these goals.

7.4.1 Multiplexing

Whenever the bandwidth of a medium linking two devices is greater than the bandwidth needs of the devices, the link can be shared. **Multiplexing** is the set of techniques that allows the simultaneous transmission of multiple signals across a single data link. As data and telecommunications use increases, so does traffic. We can accommodate this increase by continuing to add individual links each time a new channel is needed, or we can install higher-bandwidth links and use each to carry multiple signals. Today's technology includes high-bandwidth media such as optical fiber and terrestrial and satellite microwaves. Each has a bandwidth far in excess of that needed for the average transmission signal. If the bandwidth of a link is greater than the bandwidth needs of the devices connected to it, the bandwidth is wasted. An efficient system maximizes the utilization of all resources; bandwidth is one of the most precious resources we have in data communications.

In a multiplexed system, n lines share the bandwidth of one link. Figure 7.44 shows the basic format of a multiplexed system. The lines on the left direct their transmission streams to a *multiplexer*, which combines them into a single stream (many-to-one). At the receiving end, that stream is fed into a *demultiplexer*, which separates the stream back into its component transmissions (one-to-many) and directs them to

Figure 7.44 Dividing a link into channels



their corresponding lines. In the figure, the word *link* refers to the physical path. The word *channel* refers to the portion of a link that carries a transmission between a given pair of lines. One link can have many (n) channels.

There are three basic multiplexing techniques: frequency-division multiplexing (FDM), wavelength-division multiplexing (WDM), and time-division multiplexing (TDM). The first two are techniques designed for analog signals, the third, for digital signals.

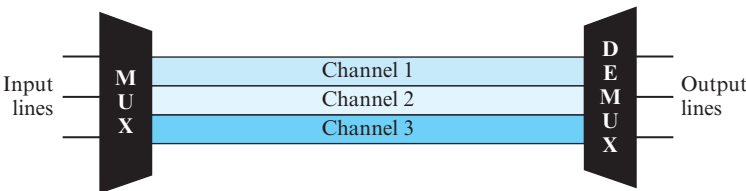
Some books use the terms FDMA and TDMA and FDM (or TDMA and TDM) interchangeably. We believe that FDMA and TDMA are access protocols defined at the data-link layer that use the service of FDM and TDM at the physical layer. We discussed FDMA and TDMA in Chapter 6. Also, some textbooks consider *code division multiple access* (CDMA) as a fourth multiplexing category. We discussed CDMA as an access method in the data-link layer in Chapter 6; it has no counterpart in the physical layer.

Frequency-Division Multiplexing

Frequency-division multiplexing (FDM) is an analog technique that can be applied when the bandwidth of a link (in hertz) is greater than the combined bandwidths of the signals to be transmitted. In FDM, signals generated by each sending device modulate different carrier frequencies. These modulated signals are then combined into a single composite signal that can be transported by the link. Carrier frequencies are separated by sufficient bandwidth to accommodate the modulated signal. These bandwidth ranges can be thought as channels through which the various signals travel. Channels can be separated by strips of unused bandwidth—**guard bands**—to prevent signals from overlapping. In addition, carrier frequencies must not interfere with the original data frequencies.

Figure 7.45 gives a conceptual view of FDM. In this illustration, the transmission path is divided into three parts, each representing a channel that carries one transmission.

Figure 7.45 Frequency-division multiplexing



We consider FDM to be an analog multiplexing technique; however, this does not mean that FDM cannot be used to combine sources sending digital signals. Digital signals can be converted to analog signals before FDM is used to multiplex them.

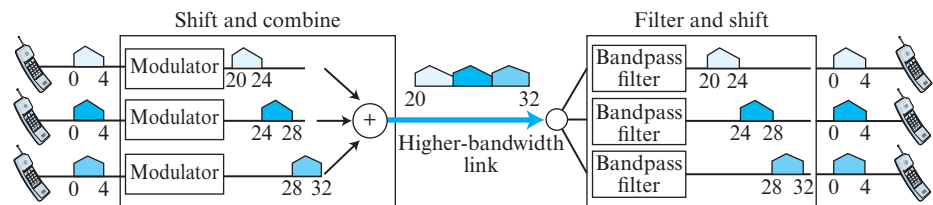
Example 7.14

Assume that a voice channel occupies a bandwidth of 4 kHz. We need to combine three voice channels into a link with a bandwidth of 12 kHz, from 20 to 32 kHz. Show the configuration, using the frequency domain. Assume there are no guard bands.

Solution

We shift (modulate) each of the three voice channels to a different bandwidth, as shown in Figure 7.46. We use the 20- to 24-kHz bandwidth for the first channel, the 24- to 28-kHz bandwidth for the second channel, and the 28- to 32-kHz bandwidth for the third one. Then we combine them as shown in Figure 7.46. At the receiver, each channel receives the entire signal, using a filter to separate out its own signal. The first channel uses a filter that passes frequencies between 20 and 24 kHz and filters out (discards) any other frequencies. The second channel uses a filter that passes frequencies between 24 and 28 kHz, and the third channel uses a filter that passes frequencies between 28 and 32 kHz. Each channel then shifts the frequency to start from zero.

Figure 7.46 Example 7.14



Wavelength-Division Multiplexing

Wavelength-division multiplexing (WDM) is designed to use the high-data-rate capability of fiber-optic cable. The optical fiber data rate is higher than the data rate of metallic transmission cable. Using a fiber-optic cable for one single line wastes the available bandwidth. Multiplexing allows us to combine several lines into one.

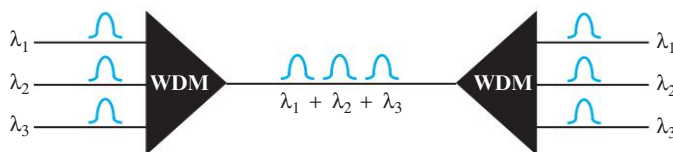
WDM is conceptually the same as FDM, except that the multiplexing and demultiplexing involve optical signals transmitted through fiber-optic channels. The idea is the same: We are combining different signals of different frequencies. The difference is that the frequencies are very high.

Figure 7.47 gives a conceptual view of a WDM multiplexer and demultiplexer. Very narrow bands of light from different sources are combined to make a wider band of light. At the receiver, the signals are separated by the demultiplexer.

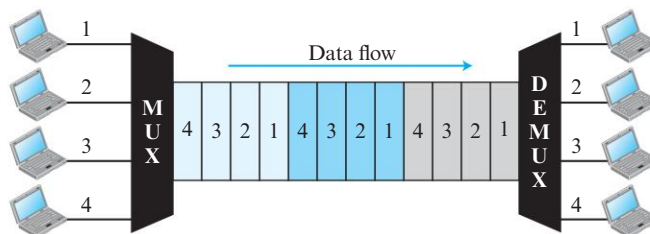
One application of WDM is the SONET network, in which multiple optical fiber lines are multiplexed and demultiplexed. We discussed SONET in Chapter 5.

Time-Division Multiplexing

Time-division multiplexing (TDM) is a digital process that allows several connections to share the high bandwidth of a link. Instead of sharing a portion of the bandwidth as in

Figure 7.47 Wavelength-division multiplexing

FDM, time is shared. Each connection occupies a portion of time in the link. Figure 7.48 gives a conceptual view of TDM. Note that the same link is used as in FDM; here, however, the link is shown sectioned by time rather than by frequency. In the figure, portions of signals 1, 2, 3, and 4 occupy the link sequentially.

Figure 7.48 TDM

Note that in Figure 7.48 we are concerned with only multiplexing, not switching. This means that all the data in a message from source 1 always go to one specific destination, be it 1, 2, 3, or 4. The delivery is fixed and unvarying, unlike switching.

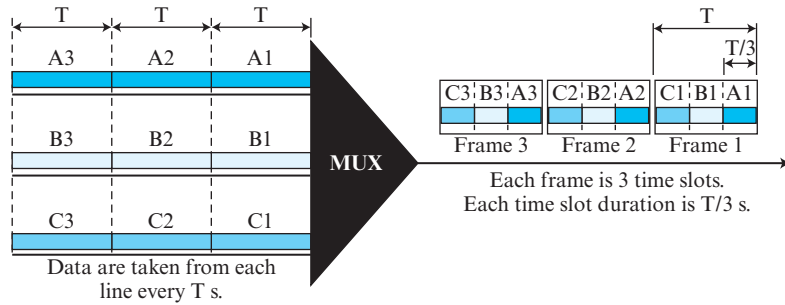
We also need to remember that TDM is, in principle, a digital multiplexing technique. Digital data from different sources are combined into one timeshared link. However, this does not mean that the sources cannot produce analog data; analog data can be sampled, changed to digital data, and then multiplexed by using TDM.

We can divide TDM into two different schemes: synchronous and statistical. We first discuss **synchronous TDM** and then show how **statistical TDM** differs. In synchronous TDM, each input connection has an allotment in the output even if it is not sending data.

Synchronous TDM

In synchronous TDM, the data flow of each input connection is divided into units, where each input occupies one input time slot. A unit can be 1 bit, one character, or one block of data. Each input unit becomes one output unit and occupies one output time slot, but the duration of an output time slot is n times shorter than the duration of an input time slot. If an input time slot is T seconds, the output time slot is T/n seconds, where n is the number of connections. In other words, a unit in the output connection has a shorter duration; it travels faster. Figure 7.49 shows an example of synchronous TDM where n is 3.

In synchronous TDM, a round of data units from each input connection is collected into a frame (we will see the reason for this shortly). If we have n connections, a frame is

Figure 7.49 Synchronous time-division multiplexing


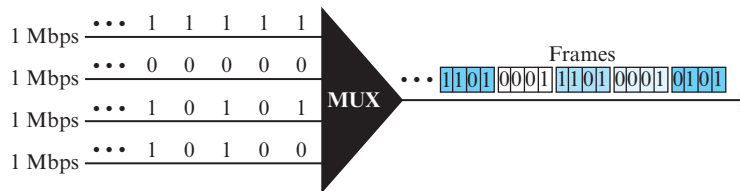
divided into n time slots and one slot is allocated for each unit, one for each input line. If the duration of the input unit is T , the duration of each slot is T/n and the duration of each frame is T (unless a frame carries some other information, as we will see shortly).

The data rate of the output link must be n times the data rate of a connection to guarantee the flow of data. In Figure 7.49, the data rate of the link is 3 times the data rate of a connection; likewise, the duration of a unit on a connection is 3 times that of the time slot (duration of a unit on the link). In the figure we represent the data prior to multiplexing as 3 times the size of the data after multiplexing. This is just to convey the idea that each unit is 3 times longer in duration before multiplexing than after.

Time slots are grouped into frames. A frame consists of one complete cycle of time slots, with one slot dedicated to each sending device. In a system with n input lines, each frame has n slots, with each slot allocated to carrying data from a specific input line.

Example 7.15

Figure 7.50 shows synchronous TDM with a data stream for each input and one data stream for the output. The unit of data is 1 bit. Find (a) the input bit duration, (b) the output bit duration, (c) the output bit rate, and (d) the output frame rate.

Figure 7.50 Example 7.15


Solution

We can answer the questions as follows:

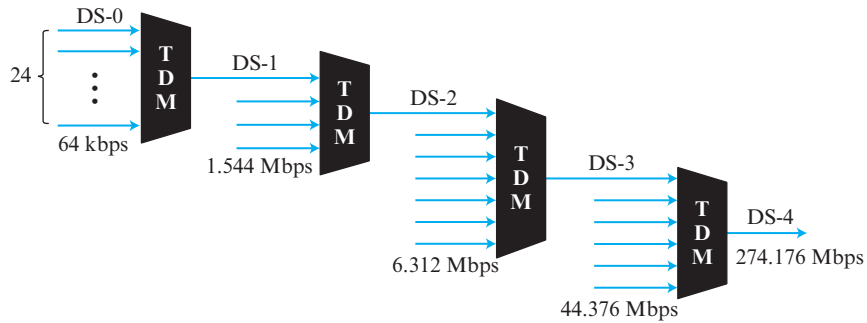
1. The input bit duration is the inverse of the bit rate: $1/1 \text{ Mbps} = 1 \mu\text{s}$.
2. The output bit duration is one-fourth of the input bit duration, or $1/4 \mu\text{s}$.
3. The output bit rate is the inverse of the output bit duration, or $1/(1/4) \mu\text{s}$ or 4 Mbps. This can also be deduced from the fact that the output rate is 4 times as fast as any input rate; so the output rate = $4 \times 1 \text{ Mbps} = 4 \text{ Mbps}$.

4. The frame rate is always the same as any input rate. So the frame rate is 1,000,000 frames per second. Because we are sending 4 bits in each frame, we can verify the result of the previous question by multiplying the frame rate by the number of bits per frame.

Example 7.16

Telephone companies implement TDM through a hierarchy of digital signals, called **digital signal (DS) service** or digital hierarchy. Figure 7.51 shows the data rates supported by each level. The commercial implementations of these services are referred to as **T lines**.

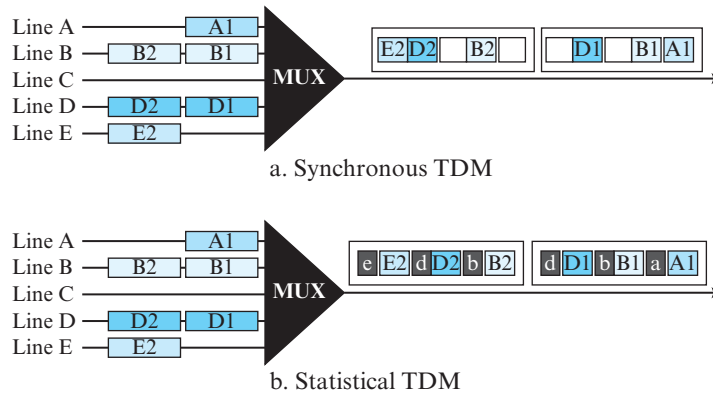
Figure 7.51 Digital hierarchy



- ❑ **DS-0** service is a single digital channel of 64 kbps.
- ❑ **DS-1** is a 1.544-Mbps service; 1.544 Mbps is 24 times 64 kbps plus 8 kbps of overhead. It can be used as a single service for 1.544-Mbps transmissions, or it can be used to multiplex 24 DS-0 channels or to carry any other combination desired by the user that can fit within its 1.544-Mbps capacity. The commercial implementation of this service is referred to as a **T-1** line.
- ❑ **DS-2** is a 6.312-Mbps service; 6.312 Mbps is 96 times 64 kbps plus 168 kbps of overhead. It can be used as a single service for 6.312-Mbps transmissions; or it can be used to multiplex 4 DS-1 channels, 96 DS-0 channels, or a combination of these service types.
- ❑ **DS-3** is a 44.376-Mbps service; 44.376 Mbps is 672 times 64 kbps plus 1.368 Mbps of overhead. It can be used as a single service for 44.376-Mbps transmissions; or it can be used to multiplex 7 DS-2 channels, 28 DS-1 channels, 672 DS-0 channels, or a combination of these service types. The commercial implementation of this service is referred to as a **T-3** line.
- ❑ **DS-4** is a 274.176-Mbps service; 274.176 Mbps is 4032 times 64 kbps plus 16.128 Mbps of overhead. It can be used to multiplex 6 DS-3 channels, 42 DS-2 channels, 168 DS-1 channels, 4032 DS-0 channels, or a combination of these service types.

Statistical Time-Division Multiplexing

As we saw in the previous section, in synchronous TDM, each input has a reserved slot in the output frame. This can be inefficient if some input lines have no data to send. In statistical time-division multiplexing, slots are dynamically allocated to improve bandwidth efficiency. Only when an input line has a slot's worth of data to send is it given a slot in the output frame. In statistical multiplexing, the number of slots in each frame is usually less than the number of input lines. The multiplexer checks each input line in round-robin fashion; it allocates a slot for an input line if the line has data to send; otherwise, it skips the line and checks the next line. Figure 7.52 shows a synchronous

Figure 7.52 TDM slot comparison

and a statistical TDM example. In the former, some slots are empty because the corresponding line does not have data to send. In the latter, however, no slot is left empty as long as there are data to be sent by any input line.

Figure 7.52 also shows a major difference between slots in synchronous TDM and statistical TDM. An output slot in synchronous TDM is totally occupied by data; in statistical TDM, a slot needs to carry data as well as the address of the destination. In synchronous TDM, there is no need for addressing; synchronization and preassigned relationships between the inputs and outputs serve as an address. We know, for example, that input 1 always goes to output 2. If the multiplexer and the demultiplexer are synchronized, this is guaranteed. In statistical multiplexing, there is no fixed relationship between the inputs and outputs because there are no preassigned or reserved slots. We need to include the address of the receiver inside each slot to show where it is to be delivered. The addressing in its simplest form can be n bits to define N different output lines with $n = \log_2 N$. For example, for eight different output lines, we need a 3-bit address.

7.4.2 Spread Spectrum

Multiplexing combines signals from several sources to achieve bandwidth efficiency; the available bandwidth of a link is divided between the sources. In **spread spectrum (SS)**, we also combine signals from different sources to fit into a larger bandwidth, but our goals are somewhat different. Spread spectrum is designed to be used in wireless applications (wireless LANs and WANs). In these types of applications, we have some concerns that outweigh bandwidth efficiency. In wireless applications, all stations use air (or a vacuum) as the medium for communication. Stations must be able to share this medium without interception by an eavesdropper and without being subject to jamming from a malicious intruder (in military operations, for example).

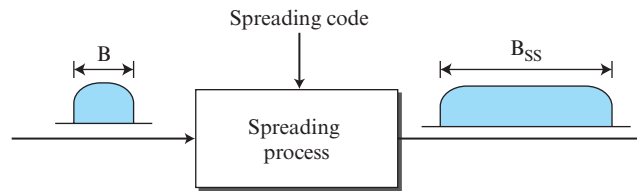
To achieve these goals, spread spectrum techniques add redundancy; they spread the original spectrum needed for each station. If the required bandwidth for each station is B , spread spectrum expands it to B_{ss} , such that $B_{ss} \gg B$. The expanded bandwidth allows the source to wrap its message in a protective envelope for a more secure transmission. An analogy is the sending of a delicate, expensive gift. We can insert the gift

in a special box to prevent it from being damaged during transportation, and we can use a superior delivery service to guarantee the safety of the package.

Figure 7.53 shows the idea of spread spectrum. Spread spectrum achieves its goals through two principles:

1. The bandwidth allocated to each station needs to be, by far, larger than what is needed. This allows redundancy.
2. The expanding of the original bandwidth B to the bandwidth B_{ss} must be done by a process that is independent of the original signal. In other words, the spreading process occurs after the signal is created by the source.

Figure 7.53 Spread spectrum



After the signal is created by the source, the spreading process uses a spreading code and spreads the bandwidth. The figure shows the original bandwidth B and the spread bandwidth B_{ss} . The spreading code is a series of numbers that look random, but are actually a pattern.

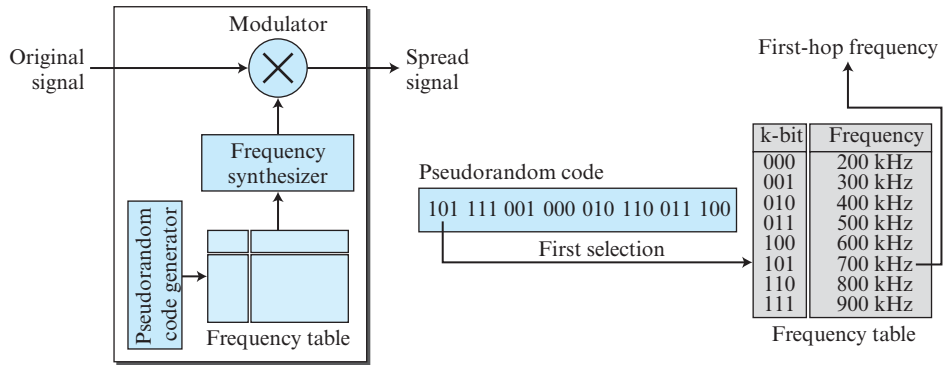
There are two techniques to spread the bandwidth: frequency hopping spread spectrum (FHSS) and direct sequence spread spectrum (DSSS).

Frequency Hopping Spread Spectrum (FHSS)

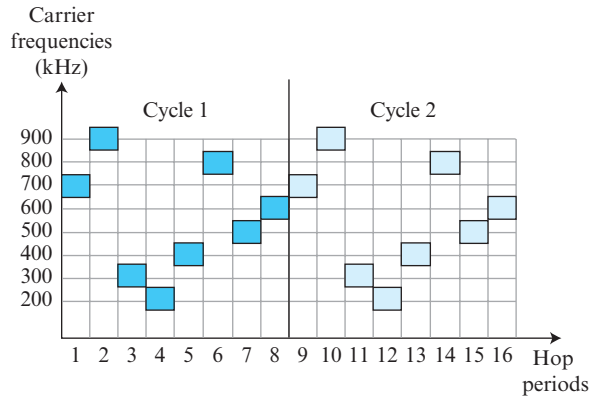
The **frequency hopping spread spectrum (FHSS)** technique uses M different carrier frequencies that are modulated by the source signal. At one moment, the signal modulates one carrier frequency; at the next moment, the signal modulates another carrier frequency. Although the modulation is done using one carrier frequency at a time, M frequencies are used in the long run. The bandwidth occupied by a source after spreading is $B_{FHSS} \gg B$. Figure 7.54 shows the general layout for FHSS.

A **pseudorandom code generator**, called **pseudorandom noise (PN)**, creates a k -bit pattern for every **hopping period** T_h . The frequency table uses the pattern to find the frequency to be used for this hopping period and passes it to the frequency synthesizer. The frequency synthesizer creates a carrier signal of that frequency, and the source signal modulates the carrier signal. Suppose we have decided to have eight hopping frequencies. This is extremely low for real applications and is just for illustration. In this case, M is 8 and k is 3. The pseudorandom code generator will create eight different 3-bit patterns. These are mapped to eight different frequencies in the frequency table.

The pattern for this station is 101, 111, 001, 000, 010, 011, 100. Note that the pattern is pseudorandom; it is repeated after eight hoppings. This means that at hopping period 1, the pattern is 101. The frequency selected is 700 kHz; the source signal modulates this carrier frequency. The second k -bit pattern selected is 111, which selects

Figure 7.54 Frequency hopping spread spectrum (FHSS)

the 900-kHz carrier; the eighth pattern is 100, the frequency is 600 kHz. After eight hoppings, the pattern repeats, starting from 101 again. Figure 7.55 shows how the signal hops around from carrier to carrier. We assume the required bandwidth of the original signal is 100 kHz.

Figure 7.55 FHSS cycles

It can be shown that this scheme can accomplish the previously mentioned goals. If there are many k -bit patterns and the hopping period is short, a sender and receiver can have privacy. If an intruder tries to intercept the transmitted signal, she can only access a small piece of data because she does not know the spreading sequence to quickly adapt herself to the next hop. The scheme also has an antijamming effect. A malicious sender may be able to send noise to jam the signal for one hopping period (randomly), but not for the whole period.

Bandwidth Sharing

If the number of hopping frequencies is M , we can multiplex M channels into one by using the same B_{ss} bandwidth. This is possible because a station uses just one

frequency in each hopping period; $M - 1$ other frequencies can be used by other $M - 1$ stations. In other words, M different stations can use the same B_{ss} if an appropriate modulation technique such as multiple FSK (MFSK) is used. FHSS is similar to FDM, as shown in Figure 7.56.

Figure 7.56 Bandwidth sharing

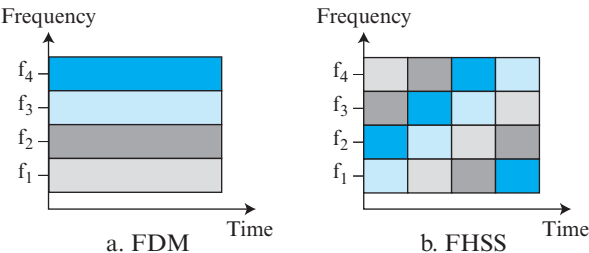
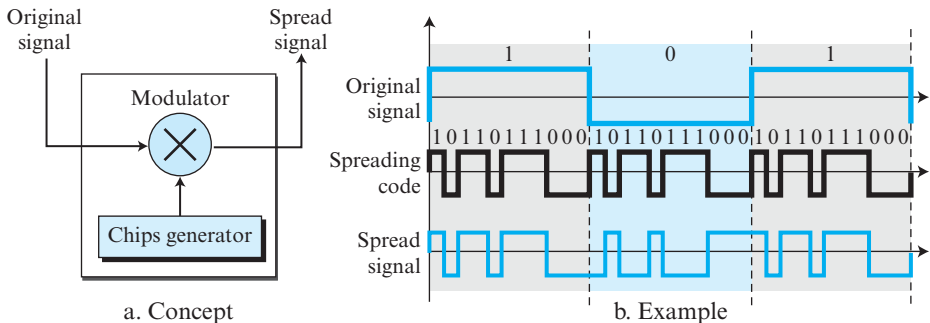


Figure 7.56 shows an example of four channels using FDM and four channels using FHSS. In FDM, each station uses $1/M$ of the bandwidth, but the allocation is fixed; in FHSS, each station uses $1/M$ of the bandwidth, but the allocation changes hop to hop.

Direct Sequence Spread Spectrum

The **direct sequence spread spectrum (DSSS)** technique also expands the bandwidth of the original signal, but the process is different. In DSSS, we replace each data bit with n bits, using a spreading code. In other words, each bit is assigned a code of n bits, called chips, where the chip rate is n times that of the data bit. Figure 7.57 shows the concept of DSSS.

Figure 7.57 DSSS



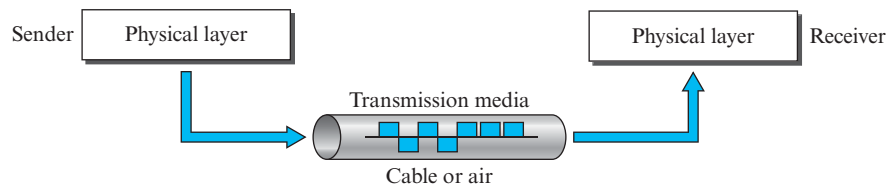
As an example, let us consider the sequence used in a wireless LAN, the famous **Barker sequence** where n is 11. We assume that the original signal and the chips in the chip generator use polar NRZ encoding. Figure 7.57 shows the chips and the result of multiplying the original data by the chips to get the spread signal.

In Figure 7.57, the spreading code is 11 chips having the pattern 10110111000 (in this case). If the original signal rate is N , the rate of the spread signal is $11N$. This means that the required bandwidth for the spread signal is 11 times larger than the bandwidth of the original signal. The spread signal can provide privacy if the intruder does not know the code. It can also provide immunity against interference if each station uses a different code.

7.5 TRANSMISSION MEDIA

We discussed many issues related to the physical layer in this chapter. In this section, we discuss transmission media. Transmission media are actually located below the physical layer and are directly controlled by the physical layer. We could say that transmission media belong to layer zero. Figure 7.58 shows the position of transmission media in relation to the physical layer.

Figure 7.58 Transmission media and physical layer



A **transmission medium** can be broadly defined as anything that can carry information from a source to a destination. For example, the transmission medium for two people having a dinner conversation is the air. The air can also be used to convey the message in a smoke signal or semaphore. For a written message, the transmission medium might be a mail carrier, a truck, or an airplane.

In data communications the definition of the information and the transmission medium is more specific. The transmission medium is usually free space, metallic cable, or fiber-optic cable. The information is usually a signal that is the result of a conversion of data from another form.

Electromagnetic energy, a combination of electric and magnetic fields vibrating in relation to each other, includes power, radio waves, infrared light, visible light, ultraviolet light, and X, gamma, and cosmic rays. Each of these constitutes a portion of the **electromagnetic spectrum**. Not all portions of the spectrum are currently usable for telecommunications, however. The media to harness those that are usable are also limited to a few types.

In telecommunications, transmission media can be divided into two broad categories: guided and unguided. Guided media include twisted-pair cable, coaxial cable, and fiber-optic cable. Unguided medium is free space.

7.5.1 Guided Media

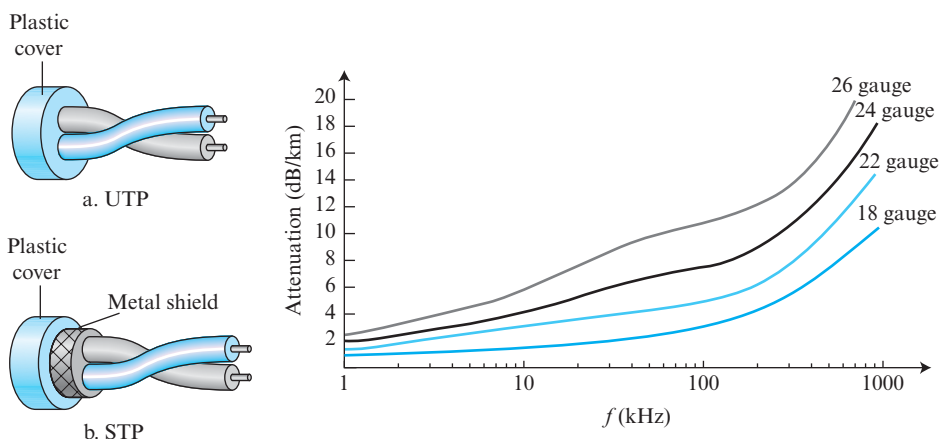
Guided media, which are those that provide a conduit from one device to another, include **twisted-pair cable**, **coaxial cable**, and **fiber-optic cable**. A signal traveling along any of these media is directed and contained by the physical limits of the

medium. Twisted-pair and coaxial cable use metallic (copper) conductors that accept and transport signals in the form of electric current. Fiber-optic cable is a cable that accepts and transports signals in the form of light.

Twisted-Pair Cable

A twisted pair consists of two conductors (normally copper), each with its own plastic insulation, twisted together, as shown in Figure 7.59.

Figure 7.59 Twisted-pair cable



One of the wires is used to carry signals to the receiver, and the other is used only as a ground reference. The receiver uses the difference between the two.

In addition to the signal from the sender, interference (noise) and crosstalk may affect both wires and create unwanted signals. If the two wires are parallel, the effect of these unwanted signals is not the same in both wires because they are at different locations relative to the noise or crosstalk sources. This results in a difference at the receiver. By twisting the pairs, a balance is maintained. The most common twisted-pair cable used in communications is referred to as **unshielded twisted-pair (UTP)**. IBM has also produced a version of twisted-pair cable for its use called **shielded twisted-pair (STP)**. STP cable has a metal foil or braided-mesh covering that encases each pair of insulated conductors. Although metal casing improves the quality of cable by preventing the penetration of noise or crosstalk, it is bulkier and more expensive.

Performance

One way to measure the performance of twisted-pair cable is to compare attenuation versus frequency and distance. A twisted-pair cable can pass a wide range of frequencies. However, Figure 7.59 also shows that with increasing frequency, the attenuation, measured in decibels per kilometer (dB/km), sharply increases with frequencies above 100 kHz. Note that **gauge** is a measure of the thickness of the wire (inversely).

Applications

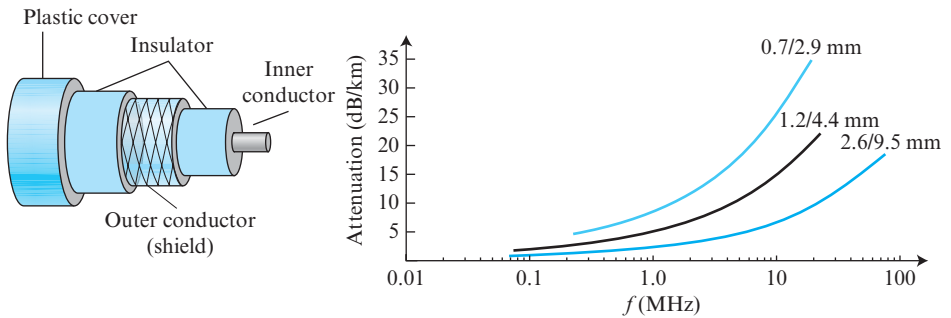
Twisted-pair cables are used in telephone lines to provide voice and data channels. The local loop—the line that connects subscribers to the central telephone office—commonly

consists of unshielded twisted-pair cables. The DSL lines that are used by the telephone companies to provide high-data-rate connections also use the high-bandwidth capability of unshielded twisted-pair cables. Local-area networks, such as 10Base-T and 100Base-T, also use twisted-pair cables. We discussed these networks in Chapter 5.

Coaxial Cable

Coaxial cable (or *coax*) carries signals of higher frequency ranges than those in twisted-pair cable, in part because the two media are constructed quite differently. Instead of having two wires, coax has a central core conductor of solid or stranded wire (usually copper) enclosed in an insulating sheath, which is, in turn, encased in an outer conductor of metal foil, braid, or a combination of the two. The outer metallic wrapping serves both as a shield against noise and as the second conductor, which completes the circuit. This outer conductor is also enclosed in an insulating sheath, and the whole cable is protected by a plastic cover (see Figure 7.60).

Figure 7.60 Coaxial cable



Performance

As we did with twisted-pair cables, we can measure the performance of a coaxial cable. We notice in Figure 7.60 that the attenuation is much higher in coaxial cable than in twisted-pair cable. In other words, although coaxial cable has a much higher bandwidth, the signal weakens rapidly and requires the frequent use of repeaters.

Applications

Coaxial cable was widely used in analog telephone networks where a single coaxial network could carry 10,000 voice signals. Later it was used in digital telephone networks where a single coaxial cable could carry digital data up to 600 Mbps. However, coaxial cable in telephone networks has largely been replaced today with fiber-optic cable.

Cable TV networks also use coaxial cable. In the traditional cable TV network, the entire network used coaxial cable. Later, however, cable TV providers replaced most of the media with fiber-optic cable; hybrid networks use coaxial cable only at the network boundaries, near the consumer premises. Cable TV uses RG-59 coaxial cable.

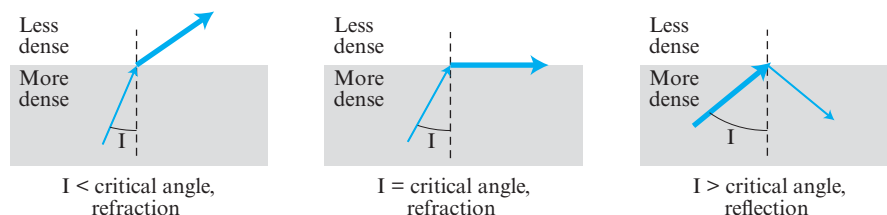
Another common application of coaxial cable is in traditional Ethernet LANs (see Chapter 5). Because of its high bandwidth, and consequently high data rate, coaxial cable was chosen for digital transmission in early Ethernet LANs. Thick Ethernet has specialized connectors.

Fiber-Optic Cable

A fiber-optic cable is made of glass or plastic and transmits signals in the form of light. To understand optical fiber, we first need to explore several aspects of the nature of light.

Light travels in a straight line as long as it is moving through a single uniform substance. If a ray of light traveling through one substance suddenly enters another substance (of a different density), the ray changes direction. Figure 7.61 shows how a ray of light changes direction when going from a more dense to a less dense substance.

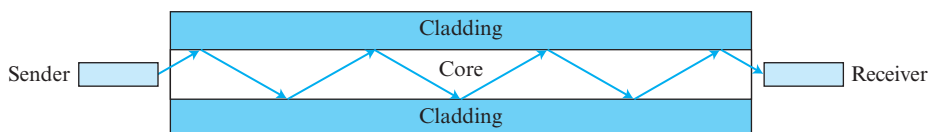
Figure 7.61 Bending of light ray



As the figure shows, if the **angle of incidence** I (the angle the ray makes with the line perpendicular to the interface between the two substances) is less than the **critical angle**, the ray **refracts** and moves closer to the surface. If the angle of incidence is equal to the critical angle, the light bends along the interface. If the angle is greater than the critical angle, the ray **reflects** (makes a turn) and travels again in the denser substance. Note that the critical angle is a property of the substance, and its value differs from one substance to another.

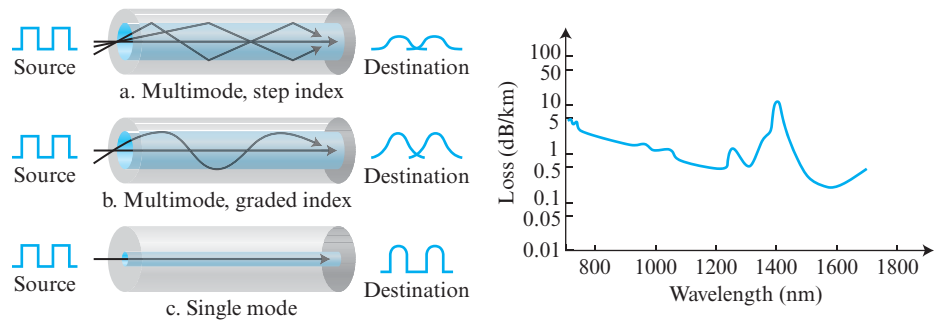
Optical fibers use reflection to guide light through a channel. A glass or plastic **core** is surrounded by a **cladding** of less dense glass or plastic. The difference in density of the two materials must be such that a beam of light moving through the core is reflected off the cladding instead of being refracted into it. See Figure 7.62.

Figure 7.62 Optical fiber



Propagation Modes

Current technology supports two modes (multimode and single mode) for propagating light along optical channels, each requiring fiber with different physical characteristics. Multimode can be implemented in two forms: step-index or graded-index. Multimode is so named because multiple beams from a light source move through the core in different paths. How these beams move within the cable depends on the structure of the core, as shown in Figure 7.63.

Figure 7.63 Modes

In **multimode step-index fiber**, the density of the core remains constant from the center to the edges. A beam of light moves through this constant density in a straight line until it reaches the interface of the core and the cladding. At the interface, there is an abrupt change due to a lower density; this alters the angle of the beam's motion. The term *step index* refers to the suddenness of this change, which contributes to the distortion of the signal as it passes through the fiber.

A second type of fiber, called **multimode graded-index fiber**, decreases this distortion of the signal through the cable. The word *index* here refers to the index of refraction. As we saw above, the index of refraction is related to density. A graded-index fiber, therefore, is one with varying densities. Density is highest at the center of the core and decreases gradually to its lowest at the edge.

Single-mode uses step-index fiber and a highly focused source of light that limits beams to a small range of angles, all close to the horizontal. The **single-mode fiber** itself is manufactured with a much smaller diameter than that of multimode fiber, and with substantially lower density (index of refraction). The decrease in density results in a critical angle that is close enough to 90° to make the propagation of beams almost horizontal. In this case, propagation of different beams is almost identical, and delays are negligible. All the beams arrive at the destination "together" and can be recombined with little distortion to the signal.

Performance

The plot of attenuation versus wavelength in Figure 7.63 also shows a very interesting phenomenon in fiber-optic cable. Attenuation is flatter than in the case of twisted-pair cable and coaxial cable. The performance is such that we need fewer (actually 10 times fewer) repeaters when we use fiber-optic cable.

Applications

Fiber-optic cable is often found in backbone networks because its wide bandwidth is cost-effective. Today, with wavelength-division multiplexing (WDM), we can transfer data at a rate of 1600 Gbps. The SONET network, which we discussed in Chapter 5, provides such a backbone.

Some cable TV companies use a combination of optical fiber and coaxial cable, thus creating a hybrid network. Optical fiber provides the backbone structure while

coaxial cable provides the connection to the user premises. This is a cost-effective configuration since the narrow bandwidth requirement at the user end does not justify the use of optical fiber.

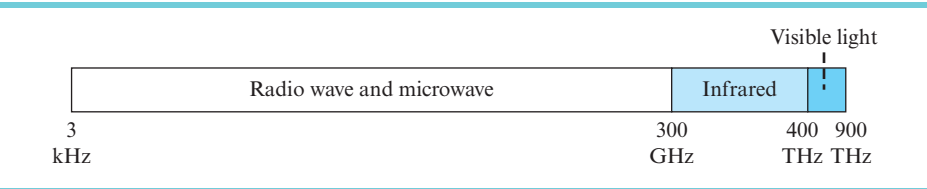
Local-area networks such as 100Base-FX network (Fast Ethernet) and 1000Base-X also use fiber-optic cable. The 10-Gigabit Ethernet operation also uses fiber-optic cable.

7.5.2 Unguided Media: Wireless

Unguided media transport electromagnetic waves without using a physical conductor. This type of communication is often referred to as *wireless communication*. Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them.

Figure 7.64 shows the part of the electromagnetic spectrum, ranging from 3 kHz to 900 THz, used for wireless communication.

Figure 7.64 Electromagnetic spectrum for wireless communication



Unguided signals can travel from source to destination in several ways: ground propagation, sky propagation, and line-of-sight propagation.

In **ground propagation**, radio waves travel through the lowest portion of the atmosphere, hugging the earth. These low-frequency signals emanate in all directions from the transmitting antenna and follow the curvature of the planet. Distance depends on the amount of power in the signal: The greater the power, the greater the distance. In **sky propagation**, higher-frequency radio waves radiate upward into the ionosphere (the layer of atmosphere where particles exist as ions) where they are reflected back to earth. This type of transmission allows for greater distances with lower output power. In **line-of-sight propagation**, very high-frequency signals are transmitted in straight lines directly from antenna to antenna. Antennas must be directional, facing each other, and either tall enough or close enough together not to be affected by the curvature of the earth. Line-of-sight propagation is tricky because radio transmissions cannot be completely focused.

The section of the electromagnetic spectrum defined as radio waves and microwaves is divided into eight ranges, called *bands*, each regulated by government authorities. These bands are rated from *very low frequency* (VLF) to *extremely high frequency* (EHF). Table 7.1 lists these bands, their ranges, propagation methods, and some applications.

We can divide wireless transmission into three broad groups: radio waves, microwaves, and infrared waves.

Radio Waves

Although there is no clear-cut demarcation between radio waves and microwaves, electromagnetic waves ranging in frequencies between 3 kHz and 1 GHz are normally

Table 7.1 Bands

<i>Band</i>	<i>Range</i>	<i>Propagation</i>	<i>Application</i>
VLF (very low frequency)	3–30 kHz	Ground	Long-range radio
LF (low frequency)	30–300 kHz	Ground	Radio beacons
MF (middle frequency)	300 kHz–3 MHz	Sky	AM radio
HF (high frequency)	3–30 MHz	Sky	Citizens band (CB), ship/aircraft communication
VHF (very high frequency)	30–300 MHz	Sky and line-of-sight	VHF TV, FM radio
UHF (ultrahigh frequency)	300 MHz–3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
SHF (superhigh frequency)	3–30 GHz	Line-of-sight	Satellite communication
EHF (extremely high frequency)	30–300 GHz	Line-of-sight	Radar, satellite

called **radio waves**; waves ranging in frequencies between 1 and 300 GHz are called **microwaves**. However, the behavior of the waves, rather than the frequencies, is a better criterion for classification. Radio waves, for the most part, are omnidirectional. When an antenna transmits radio waves, they are propagated in all directions. This means that the sending and receiving antennas do not have to be aligned. A sending antenna sends waves that can be received by any receiving antenna. The omnidirectional property has a disadvantage, too. The radio waves transmitted by one antenna are susceptible to interference by another antenna that may send signals using the same frequency or band.

Radio waves, particularly those waves that propagate in the sky mode, can travel long distances. This makes radio waves a good candidate for long-distance broadcasting such as AM radio.

Radio waves, particularly those of low and medium frequencies, can penetrate walls. This characteristic can be both an advantage and a disadvantage. It is an advantage because, for example, an AM radio can receive signals inside a building. It is a disadvantage because we cannot isolate a communication to just inside or outside a building. The radio wave band is relatively narrow, just under 1 GHz, compared to the microwave band. When this band is divided into subbands, the subbands are also narrow, leading to a low data rate for digital communications.

Almost the entire band is regulated by authorities (e.g., the FCC in the United States). Using any part of the band requires permission from the authorities.

Microwaves

Electromagnetic waves having frequencies between 1 and 300 GHz are called microwaves. Microwaves are unidirectional. When an antenna transmits microwaves, they can be narrowly focused. This means that the sending and receiving antennas need to be aligned. The unidirectional property has an obvious advantage. A pair of antennas

can be aligned without interfering with another pair of aligned antennas. The following describes some characteristics of microwave propagation:

- ❑ Microwave propagation is line-of-sight. Since the towers with the mounted antennas need to be in direct sight of each other, towers that are far apart need to be very tall. The curvature of the earth as well as other blocking obstacles do not allow two short towers to communicate by using microwaves. Repeaters are often needed for long-distance communication.
- ❑ Very high-frequency microwaves cannot penetrate walls. This characteristic can be a disadvantage if receivers are inside buildings.
- ❑ The microwave band is relatively wide, almost 299 GHz. Therefore wider subbands can be assigned, and a high data rate is possible.
- ❑ Use of certain portions of the band requires permission from authorities.

Applications

Microwaves, due to their unidirectional properties, are very useful when unicast (one-to-one) communication is needed between the sender and the receiver. They are used in cellular phones, satellite networks, and wireless LANs.

Infrared

Infrared waves, with frequencies from 300 GHz to 400 THz (wavelengths from 1 mm to 770 nm), can be used for short-range communication. Infrared waves, having high frequencies, cannot penetrate walls. This advantageous characteristic prevents interference between one system and another; a short-range communication system in one room cannot be affected by another system in the next room. When we use our infrared remote control, we do not interfere with the use of the remote by our neighbors. However, this same characteristic makes infrared signals useless for long-range communication. In addition, we cannot use infrared waves outside a building because the sun's rays contain infrared waves that can interfere with the communication.

7.6 END-CHAPTER MATERIALS

7.6.1 Recommended Reading

For more details about subjects discussed in this chapter, we recommend the following books. The items in brackets [...] refer to the reference list at the end of the text.

Books

Several books cover topics discussed in this chapter including [Pea 92], [Cou 01], [Ber 96], [Hsu 03], [Spi 74], [Sta 04], [Tan 03], [G & W 04], [SSS 05], [BEL 01], and [Max 99].

7.6.2 Key Terms

alternate mark inversion (AMI)
 amplitude modulation (AM)
 amplitude shift keying (ASK)
 analog data

analog signal
 analog-to-analog conversion
 analog-to-digital conversion
 angle of incidence

attenuation	multimode graded-index fiber
band-pass channel	multimode step-index fiber
bandwidth	multiplexing
Barker sequence	noise
baseband transmission	nonperiodic signal
bipolar encoding	nonreturn to zero (NRZ)
bipolar with 8-zero substitution (B8ZS)	Nyquist bit rate
bit rate	Nyquist theorem
coaxial cable	peak amplitude
composite signal	period
constellation diagram	periodic signal
critical angle	phase
crosstalk	phase modulation (PM)
decibel (dB)	phase shift keying (PSK)
delta modulation	pseudorandom noise (PN)
differential Manchester	pseudoternary
digital data	pulse amplitude modulation (PAM)
digital data service (DDS)	pulse code modulation (PCM)
digital signal	quadrature amplitude modulation (QAM)
digital signal (DS) service	radio wave
digital-to-analog conversion	return-to-zero (RZ)
digital-to-digital conversion	sample and hold
digitization	sampling
direct sequence spread spectrum	sampling rate
(DSSS) distortion	scrambling
eight-binary, six-ternary (8B6T)	Shannon capacity
electromagnetic spectrum	shielded twisted-pair (STP)
fiber-optic cable	signal-to-noise ratio (SNR)
four binary/five binary (4B/5B)	sine wave
frequency	single-mode fiber
frequency-division multiplexing (FDM)	sky propagation
frequency-domain plot	spread spectrum (SS)
frequency-hopping spread spectrum	statistical TDM
frequency modulation (FM)	synchronous TDM
frequency shift keying (FSK)	T line
ground propagation	throughput
guard band	time-division multiplexing (TDM)
guided media	time-domain plot
high-density bipolar 3-zero (HDB3)	transmission medium
jitter	twisted-pair cable
line coding	two-binary, one quaternary (2B1Q)
line-of-sight propagation	unguided media
low-pass channel	unshielded twisted-pair (UTP)
Manchester	wavelength-division multiplexing (WDM)

7.6.3 Summary

Data must be transformed to electromagnetic signals to be transmitted. Analog data are continuous and take continuous values. Digital data have discrete states and take discrete values. Analog signals can have an infinite number of values in a range; digital signals

can have only a limited number of values. In data communications, we commonly use periodic analog signals and nonperiodic digital signals.

Digital-to-digital conversion involves three techniques: line coding, block coding, and scrambling. Line coding is the process of converting digital data to a digital signal. Block coding provides redundancy to ensure synchronization and inherent error detection. The most common technique to change an analog signal to digital data (digitization) is called pulse code modulation (PCM).

Digital-to-analog conversion is the process of changing one of the characteristics of an analog signal based on the information in the digital data. Digital-to-analog conversion can be accomplished in several ways: amplitude shift keying (ASK), frequency shift keying (FSK), and phase shift keying (PSK). Quadrature amplitude modulation (QAM) combines ASK and PSK. Analog-to-analog conversion can be accomplished in three ways: amplitude modulation (AM), frequency modulation (FM), and phase modulation (PM).

Bandwidth utilization is the use of available bandwidth to achieve specific goals. Efficiency can be achieved by using multiplexing; privacy and antijamming can be achieved by using spreading.

Transmission media lie below the physical layer. A guided medium provides a physical conduit from one device to another. Twisted-pair cable, coaxial cable, and optical fiber are the most popular types of guided media. Unguided media (free space) transport electromagnetic waves without the use of a physical conductor.

7.7 PRACTICE SET

7.7.1 Quizzes

A set of interactive quizzes for this chapter can be found on the book website. It is strongly recommended that students take the quizzes to check his/her understanding of the materials before continuing with the practice set.

7.7.2 Questions

- Q7-1.** Which of the following is a cause of transmission impairment?
 a. attenuation b. modulation c. noise
- Q7-2.** Which of the following is the characteristic of the low-pass channel?
 a. A channel with a bandwidth that starts from zero.
 b. A channel with a bandwidth that does not start from zero.
- Q7-3.** We send a voice signal from a microphone to a recorder. Is this baseband or broadband transmission?
- Q7-4.** We send a digital signal from one station on a LAN to another station. Is this baseband or broadband transmission?
- Q7-5.** How can we find the period of a sine wave when its frequency is given?
- Q7-6.** Which of the following, measures the value of a signal at any time?
 a. amplitude b. frequency c. phase
- Q7-7.** Can we say whether a signal is periodic or nonperiodic by just looking at its time domain plot? If so, how?
- Q7-8.** Is the frequency domain plot of a voice signal discrete or continuous?

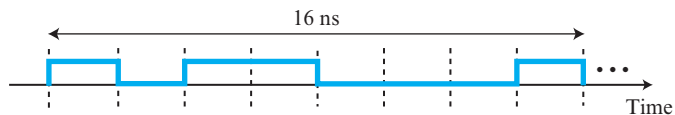
- Q7-9.** Which of the following defines the theoretical maximum bit rate of a noiseless channel?
a. Nyquist theorem **b.** Shannon capacity
- Q7-10.** Which of the following techniques is an example of digital-to-digital conversion?
a. line coding **b.** block coding **c.** amplitude modulation
- Q7-11.** Define block coding and give its purpose.
- Q7-12.** Describe PCM.
- Q7-13.** Which of the following is the definition of a baseband transmission?
a. Sending a digital or an analog signal without modulation using a low-pass channel.
b. Modulating a digital or an analog signal using a band-pass channel.
- Q7-14.** How can a periodic composite signal be decomposed into its individual frequencies?
- Q7-15.** Which of the two digital-to-analog conversion techniques is more susceptible to noise?
a. ASK **b.** FSK
- Q7-16.** Which component of a signal is shown on the horizontal axis in a constellation diagram?
- Q7-17.** Define analog transmission.
- Q7-18.** Which characteristics of an analog signal are changed to represent the digital signal in each of the following digital-to-analog conversion mechanisms?
a. ASK **b.** PSK
- Q7-19.** Define a link in multiplexing.
- Q7-20.** Which of the three multiplexing techniques are used to combine analog signals?
- Q7-21.** Define synchronous TDM and compare it with statistical TDM.
- Q7-22.** Define FHSS and explain how it achieves bandwidth spreading.
- Q7-23.** Which characteristics of an analog signal are changed to represent the low-pass analog signal in each of the following analog-to-analog conversions?
a. FM **b.** PM
- Q7-24.** Describe multiplexing.
- Q7-25.** What are the three major classes of guided media?
- Q7-26.** What is the purpose of cladding in an optical fiber?
- Q7-27.** Describe how waves in sky propagation travel from the source to the destination.
- Q7-28.** Describe how omnidirectional waves are propagated.
- Q7-29.** What is the position of the transmission media in the TCP/IP protocol suite?
- Q7-30.** Name the two major categories of transmission media.

7.7.3 Problems

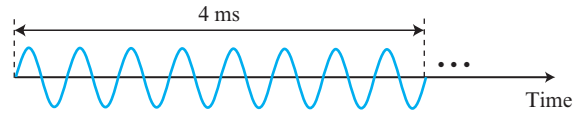
- P7-1.** What is the bandwidth of a signal that can be decomposed into five sine waves with frequencies at 0, 20, 50, 100, and 200 Hz? All peak amplitudes are the same. Draw signals in the frequency domain and show the bandwidth.

- P7-2.** A periodic composite signal with a bandwidth of 2000 Hz is composed of two sine waves. The first one has a frequency of 100 Hz with a maximum amplitude of 20 V; the second one has a maximum amplitude of 5 V. Draw the signals and show the bandwidth.
- P7-3.** Which signal has a wider bandwidth, a sine wave with a frequency of 100 Hz or a sine wave with a frequency of 200 Hz?
- P7-4.** What is the bit rate for each of the following signals?
- A signal in which 1-bit lasts 0.001 s
 - A signal in which 1-bit lasts 2 ms
 - A signal in which 10-bits last 20 μ s
- P7-5.** Given the frequencies listed below, calculate the corresponding periods.
- 24 Hz
 - 8 MHz
 - 140 kHz
- P7-6.** What is the phase shift for the following waves?
- A sine wave with maximum amplitude at time zero
 - A sine wave with maximum amplitude after 1/4 cycle
 - A sine wave with zero amplitude after 3/4 cycle and increasing
- P7-7.** A periodic composite signal contains frequencies from 10 to 30 kHz, each with an amplitude of 10 V. Draw the frequency spectrum and calculate the bandwidth.
- P7-8.** A nonperiodic composite signal contains frequencies from 10 to 30 kHz. The peak amplitude is 10 V for the lowest and the highest signals and is 30 V for the 20 kHz signal. Assuming that the amplitudes change gradually from the minimum to the maximum and from maximum to minimum, draw the frequency spectrum.
- P7-9.** A signal travels from point A to point B. At point A, the signal power is 100 W. At point B, the power is 90 W. What is the attenuation in decibels?
- P7-10.** The attenuation of a signal is -10 dB. What is the final signal power if it was originally 5 W?
- P7-11.** A device is sending out data at the rate of 1000 bps.
- How long does it take to send out 10-bits?
 - How long does it take to send out a character (8-bits)?
 - How long does it take to send a file of 100,000 characters?
- P7-12.** What is the bit rate for the signal in Figure 7.65?

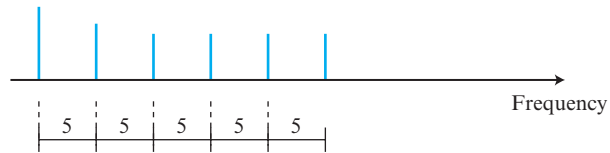
Figure 7.65 Problem P7-12



- P7-13.** What is the frequency of the signal in Figure 7.66?

Figure 7.66 Problem P7-13

- P7-14.** What is the bandwidth of the composite signal shown in Figure 7.67?

Figure 7.67 Problem P7-14

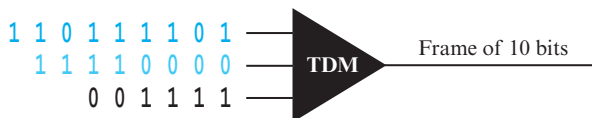
- P7-15.** We measure the performance of a telephone line (4 kHz of bandwidth). When the signal is 10 V, the noise is 10 mV. What is the maximum data rate supported by this telephone line?
- P7-16.** A file contains 2 million bytes. How long does it take to download this file using a 56 Kbps channel (dial-up modem)?
- P7-17.** A signal has passed through three cascaded amplifiers, each with a 4 dB gain. What is the total gain? How much is the signal amplified?
- P7-18.** If the bandwidth of a channel is 5 Kbps, how long does it take to send a frame of 100,000 bits into this channel?
- P7-19.** A signal has a wavelength of 1 μm in air. How far can the front of the wave travel during 1000 cycles?
- P7-20.** A line has a signal-to-noise ratio of 1000 and a bandwidth of 4 kHz. What is the maximum data rate supported by this line?
- P7-21.** If the peak voltage value of a signal is 20 times the peak voltage value of the noise, what is the SNR? What is the SNR_{dB} ?
- P7-22.** We need to upgrade a channel to a higher bandwidth. Answer the following questions:
- How will the rate (capacity) improve if we double the bandwidth?
 - How will the rate (capacity) improve if we double the SNR?
- P7-23.** A computer monitor has a resolution of 1200 by 1000 pixels. If each pixel uses 1024 color levels, how many bits are needed to send the complete contents of a screen?
- P7-24.** A signal with 200 milliwatts power passes through 10 devices, each with an average noise of 2 microwatts. What is the SNR? What is the SNR_{dB} ?
- P7-25.** In a digital transmission, the sender clock is 0.2 percent faster than the receiver clock. How many extra bits per second does the sender send if the data rate is 1 Mbps?

- P7-26.** Draw the graph of the NRZ-L scheme using each of the following data streams, assuming that the last signal level has been positive. From the graphs, guess the bandwidth for this scheme.
 a. 00000000 b. 11111111 c. 01010101 d. 00110011
- P7-27.** Draw the graph of the Manchester scheme using each of the following data streams, assuming that the last signal level has been positive. From the graphs, guess the bandwidth for this scheme.
 a. 00000000 b. 11111111 c. 01010101 d. 00110011
- P7-28.** How many invalid (unused) code sequences can we have in 5B/6B encoding?
- P7-29.** What is the transmission time of a packet sent by a station if the length of the packet is 1 million bytes (byte = 8-bits) and the bandwidth of the channel is 200 Kbps?
- P7-30.** What is the length of a bit in a channel if the propagation speed in the medium is 2×10^8 m/s and the channel bandwidth is
 a. 1 Mbps? b. 10 Mbps?
- P7-31.** How many bits can fit on a link with a 2 ms delay if the bandwidth of the link is
 a. 1 Mbps? b. 10 Mbps?
- P7-32.** What is the total delay (latency) for a frame of size 5 million bits that is being sent on a link with 10 routers each having a queuing time of 2 μ s and a processing time of 1 μ s. The length of the link is 2000 Km. The speed of propagation inside the link is 2×10^8 m/s. The link has a bandwidth of 5 Mbps. Which component of the total delay is dominant? Which one is negligible?
- P7-33.** An analog signal has a bandwidth of 4 kHz. If we sample this signal and send it through a 30 Kbps channel, what is the SNR_{dB} ?
- P7-34.** We have a baseband channel with a 1 MHz bandwidth. What is the data rate for this channel if we use each of the following line coding schemes?
 a. NRZ-L b. Manchester
- P7-35.** Calculate the baud for the given bit rate and type of modulation.
 a. 2000 bps, FSK b. 4000 bps, ASK c. 36,000 bps, 64-QAM
- P7-36.** Calculate the bit rate for the given baud and type of modulation.
 1000 baud, FSK 1000 baud, ASK 1000 baud, 16-QAM
- P7-37.** What is the number of bits per baud for the following techniques?
 a. FSK with 8 frequencies b. QAM with a 128-point constellation
- P7-38.** Draw the constellation diagram for the following:
 a. ASK, with peak amplitude values of 1 and 3
 b. 8-QAM with two different peak amplitude values, 1 and 3, and four different phases.
- P7-39.** What is the result of scrambling the sequence 11100000000000 using B8ZS scrambling techniques? Assume that the last nonzero signal level has been positive.
- P7-40.** What is the Nyquist sampling rate for each of the following signals?

- a. A low-pass signal with bandwidth of 200 kHz?
 - b. A band-pass signal with bandwidth of 200 kHz if the lowest frequency is 100 kHz?
- P7-41.** We have sampled a low-pass signal with a bandwidth of 200 kHz using 1024 levels of quantization.
- a. Calculate the bit rate of the digitized signal.
 - b. Calculate the SNR_{dB} for this signal.
 - c. Calculate the PCM bandwidth of this signal.
- P7-42.** What is the maximum data rate of a channel with a bandwidth of 200 kHz if we use four levels of digital signaling?
- P7-43.** How many bits per baud can we send in each of the following cases if the signal constellation has the given number of points?
- a. 2
 - b. 4
 - c. 16
 - d. 1024
- P7-44.** What is the required bandwidth for the following cases if we need to send 4000 bps? Let $d = 1$.
- a. ASK
 - b. FSK ($2\Delta f = 4 \text{ kHz}$)
 - c. 16-QAM
- P7-45.** A corporation has a medium with a 1 MHz bandwidth (low-pass). The corporation needs to create 10 separate independent channels, each capable of sending at least 10 Mbps. The company has decided to use QAM technology. What is the minimum number of bits per baud for each channel? What is the number of points in the constellation diagram for each channel? Let $d = 0$.
- P7-46.** Find the bandwidth to modulate a 5 kHz voice using one of the following techniques.
- a. AM
 - b. FM ($\beta = 5$)
 - c. PM ($\beta = 1$)
- P7-47.** As an example of the Nyquist theorem, let us sample a simple sine wave at three sampling rates: $f_s = 4f$ (2 times the Nyquist rate), $f_s = 2f$ (Nyquist rate), and $f_s = f$ (one half of the Nyquist rate). Show how we can recover the wave.
- P7-48.** Assume that a voice channel occupies a bandwidth of 4 kHz. We need to multiplex 10 voice channels with guard bands of 500 Hz using FDM. Calculate the required bandwidth.
- P7-49.** We have 14 sources, each creating 500 8-bit characters per second. Since only some of these sources are active at any moment, we use statistical TDM to combine these sources using character interleaving. Each frame carries 6 slots at a time, but we need to add 4-bit addresses to each slot. Answer the following questions:
- a. What is the size of an output frame in bits?
 - b. What is the output frame rate?
 - c. What is the duration of an output frame?
 - d. What is the output data rate?
- P7-50.** Show the contents of the five output frames for a synchronous TDM multiplexer that combines four sources sending the following characters. Note that the characters are sent in the same order that they are typed. The third source is silent.
- a. Source 1 message: HELLO

- b. Source 2 message: HI
 - c. Source 3 message:
 - d. Source 4 message: BYE
- P7-51.** We need to transmit 100 digitized voice channels using a pass-band channel of 20 kHz. What should be the ratio of bits/Hz if we use no guard band?
- P7-52.** We need to use synchronous TDM and combine 20 digital sources, each of 100 Kbps. Each output slot carries 1 bit from each digital source, but one extra bit is added to each frame for synchronization. Answer the following questions:
- a. What is the size of an output frame in bits?
 - b. What is the output frame rate?
 - c. What is the duration of an output frame?
 - d. What is the output data rate?
 - e. What is the efficiency of the system (ratio of useful bits to total bits)?
- P7-53.** We have a digital medium with a data rate of 10 Mbps. How many 64 kbps voice channels can be carried by this medium if we use DSSS with the Barker sequence?
- P7-54.** Assume that the power of a noiseless signal is 10 mW. The signal passes through five stages of an amplifier. Each stage doubles the power, but it also adds 10 μ W of noise to the signal. What is the SNR_{dB} of the signal after it leaves the last stage?
- P7-55.** What is the SNR_{dB} when we quantize a human voice using 8-bits per sample?
- P7-56.** We have a 5 kHz baseband channel. We need to send data at the rate of 40 Kbps using this channel.
- a. What should the number of bit levels be to achieve this rate?
 - b. How should the quality of the channel (SNR) be improved to achieve this rate?
- P7-57.** Figure 7.68 shows a synchronous TDM. Each output frame is only 10-bits long (three bits taken from each input plus one framing bit). What is the output stream?

Figure 7.68 Problem P7-57



- P7-58.** An FHSS system uses a 4-bit PN sequence. If the bit rate of the PN is 64-bits per second, answer the following questions:
- a. What is the total number of possible hops?
 - b. What is the time needed to finish a complete cycle of PN?
- P7-59.** An STP cable has a loss of 1 dB per Km at 10 kHz. We want to have a link of 10 Km using this cable. What should the power of the signal be at the source if we want the signal to have 10 mW power at the destination?